


CIRCUITI IN REGIME SINUSOIDALE E FASORI

$$\begin{aligned} v(t) &= V_m \sin(\omega t + \vartheta) & \longrightarrow & \quad v(t) = \frac{V_m e^{j(\omega t + \vartheta)} - V_m e^{-j(\omega t + \vartheta)}}{2j} \\ i(t) &= I_m \sin(\omega t + \beta) & & \quad i(t) = \frac{I_m e^{j(\omega t + \beta)} - I_m e^{-j(\omega t + \beta)}}{2j} \end{aligned}$$

FASORI

$$\bar{V} = \frac{V_m e^{j\vartheta}}{2j} \quad \bar{I} = \frac{I_m e^{j\beta}}{2j}$$

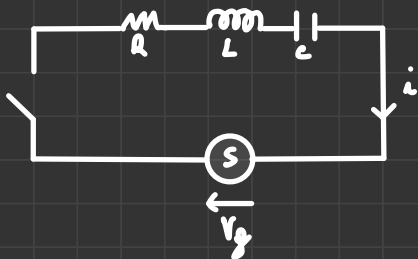
$$v(t) = \bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}$$

$$i(t) = \bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}$$

$$\text{PER } \bar{V} = \frac{V_m e^{j\vartheta}}{2j} \quad \bar{I} = \frac{I_m e^{j\beta}}{2j}$$

$$v(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j} \quad i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j}$$

FASORI NEI CIRCUITI RLC CON FORZANTE SINUSOIDALE



$$V_g(t) = V_m \sin(\omega t + \varphi)$$

$$\begin{cases} V_g(t) = R i(t) + L \frac{di(t)}{dt} + V_C(t) \\ i(t) = C \frac{dV_C(t)}{dt} \end{cases}$$

$$V_{gp}(t) = V_m \sin(\omega t + \theta) \quad V_{gp}(0) = V_m \sin(\theta)$$

$$i(t) = I_m \sin(\omega t + \beta)$$

$$V_{Cp}(t) = \frac{1}{C} \int_0^t i(t) dt + V_C(0)$$

$$V(t) = \frac{\bar{V} e^{j\omega t} - \bar{V} e^{-j\omega t}}{2j} \quad i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I} e^{-j\omega t}}{2j}$$

$$\bar{V}_g = R \bar{I} + j\omega L \bar{I} + \frac{1}{j\omega C} \bar{I} = (R + j(\omega L - \frac{1}{\omega C})) \bar{I}$$

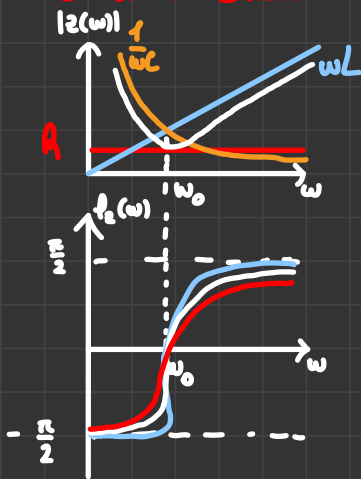
$$\bar{V} = Z \bar{I}$$

STUDIO L'IMPIEDENZA AL VARIARE DI ω NEI CIRCUITI RLC SERIE

$$Z(\omega) = R + j(\omega L - \frac{1}{\omega C})$$

$$|Z(\omega)| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

$$\varphi_z(\omega) = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$



$\omega < \omega_0$ OHMICO-CAPACITIVO
 PER $R=0$ PURAM. CAPACITIVO
 $\omega > \omega_0$ OHMICO-INDUTTIVO
 PER $R=0$ PURAM. INDUTTIVO

$\omega \rightarrow \infty$ $\omega L \gg \frac{1}{\omega C}$

$\omega = \omega_0$ $\varphi_z(\omega) = 0$

$\omega \rightarrow 0$ $\frac{1}{\omega C} \gg \omega L$

POTENZA IN REGIME PERMANENTE SINUSOIDALE

$$v(t) = V_M \sin(\omega t + \gamma) \longrightarrow v(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j}$$

$$i(t) = I_M \sin(\omega t + \beta) \quad i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j}$$

$$p(t) = v(t) i(t) = \left(\frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j} \right) \left(\frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j} \right) =$$

$$-\frac{1}{4} \left(\bar{V} \bar{I} e^{2j\omega t} - \bar{V} \bar{I}^* - \bar{V}^* \bar{I} + \bar{V}^* \bar{I}^* e^{-2j\omega t} \right)$$

$$= -\frac{1}{2} \underbrace{\frac{\bar{V} \bar{I} e^{2j\omega t} + \bar{V}^* \bar{I}^* e^{-2j\omega t}}{2}}_{\text{VARI NEL TEMPO}} + \frac{1}{2} \underbrace{\frac{\bar{V} \bar{I}^* + \bar{V}^* \bar{I}}{2}}_{\text{COSTANTE}}$$

SOSTITUZIONI

$$\bar{V} \bar{I} = V_M I_M e^{j(\gamma + \beta)} \quad \bar{V} \bar{I}^* = V_M I_M e^{j(\gamma - \beta)}$$

$$(\bar{V} \bar{I})^* = V_M I_M e^{-j(\gamma + \beta)} \quad \bar{V}^* \bar{I} = V_M I_M e^{-j(\gamma - \beta)}$$

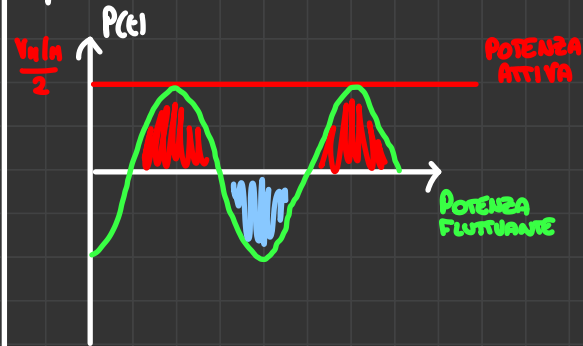
$$\gamma - \beta = \varphi$$

$$p(t) = -\frac{1}{2} V_M I_M \frac{e^{j(2\omega t + 2\gamma - \varphi)} + e^{-j(2\omega t + 2\gamma - \varphi)}}{2} + \frac{1}{2} V_M I_M \frac{e^{j\varphi} + e^{-j\varphi}}{2}$$

$$= -\frac{1}{2} V_M I_M \cos(2\omega t + 2\gamma - \varphi) + \frac{1}{2} V_M I_M \cos(\varphi)$$

$$\begin{cases} V_{\text{EFF}} = \frac{V_M}{\sqrt{2}} \longrightarrow V_M = \sqrt{2} V_{\text{EFF}} \\ I_{\text{EFF}} = \frac{I_M}{\sqrt{2}} \longrightarrow I_M = \sqrt{2} I_{\text{EFF}} \end{cases}$$

$$p(t) = V_{\text{EFF}} I_{\text{EFF}} \cos(\varphi) - V_{\text{EFF}} I_{\text{EFF}} \cos(2\omega t + 2\gamma - \varphi)$$



$$P_c = \bar{V} \bar{I}^* \quad P_c = P + Q$$

TEOREMA DEL MASSIMO TRASFERIMENTO DI POTENZA PER RETI DINAMICHE



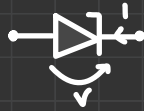
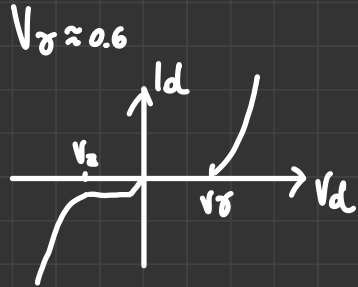
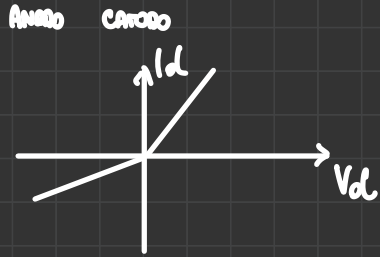
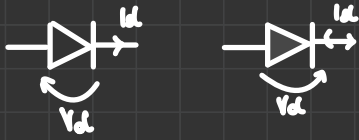
$$\bar{I} = \frac{\bar{E}_{TH}}{Z_{eq} + Z_c} = \frac{\bar{E}_{TH}}{(R_{eq} + R_c) + j(X_{eq} + X_c)}$$

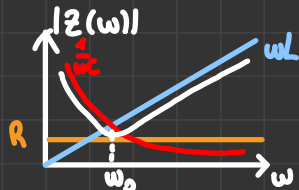
$$\dot{P}_c = Z \bar{I}^2 = R \bar{I}^2 + jX \bar{I}^2$$

$$P_c = R_c \bar{I}^2 = R_c \frac{\bar{E}_{TH}^2}{(R_{eq} + R_c)^2 + (X_{eq} + X_c)^2} \quad X_c = -X_{eq} \quad X_{TOT} = 0$$

$$Z_c = Z_{eq}^*$$

DIODI A GIUNZIONE



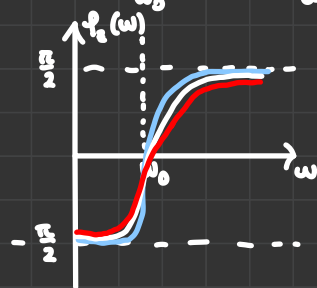


$$\omega < \omega_0 \quad 0 - c$$

$$R = 0 \quad c$$

$$\omega > \omega_0 \quad 0 - 1$$

$$R = 0 \quad 1$$

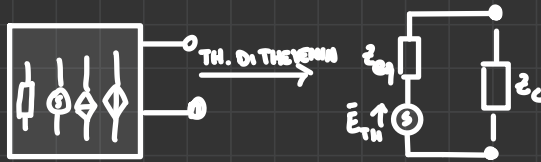


$$\omega \rightarrow \infty \quad \omega L \gg \frac{1}{\omega C} \quad \varphi_2(\omega) \rightarrow \frac{\pi}{2}$$

$$\omega = \omega_0 \quad \varphi_2(\omega) = 0$$

$$\omega \rightarrow 0 \quad \frac{1}{\omega C} \gg \omega L \quad \varphi_2(\omega) \rightarrow -\frac{\pi}{2}$$

TEOREMA DEL MASSIMO TRASFERIMENTO DI POTENZA PER ACN DINAMICHE



$$\bar{I} = \frac{\bar{E}_{TH}}{Z_{eq} + Z_C} = \frac{\bar{E}_{TH}}{(R_{eq} + R_C) + j(X_{eq} + X_C)}$$

$$\dot{P}_C = Z \cdot \bar{I}^2 = R \bar{I}^2 + jX \bar{I}^2$$

$$P_C = R_C \frac{\bar{E}_{TH}^2}{(R_{eq} + R_C)^2 + (X_{eq} + X_C)^2}$$

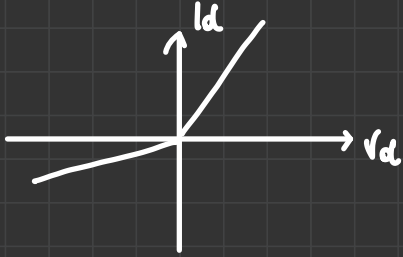
$$X_C = -X_{eq} \rightarrow X_{TOT} = 0$$

$$Z_C = Z_{eq}^*$$

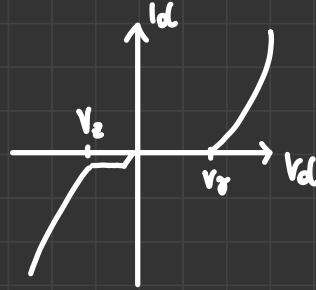
D1001



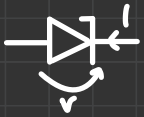
ANALOG COMPO



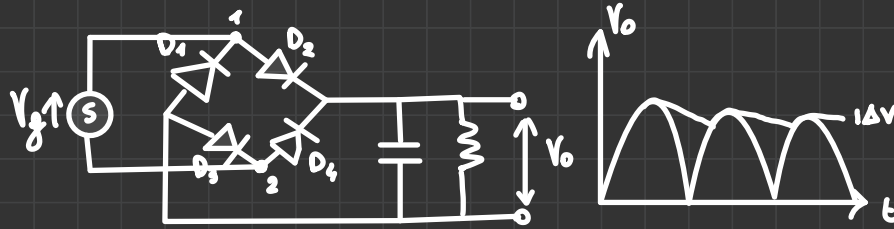
$$V_g \approx 0.6$$



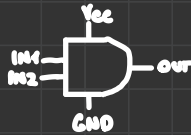
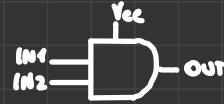
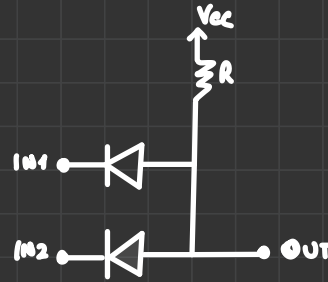
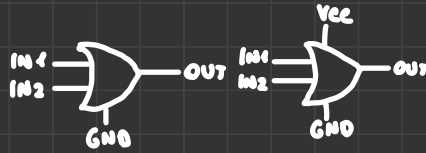
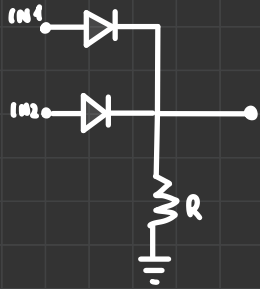
D1000 ZENER



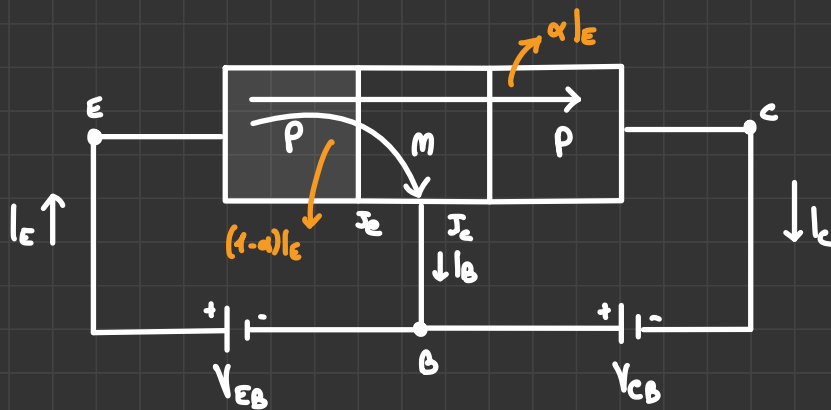
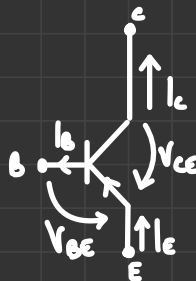
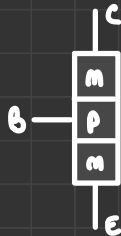
PONTE DI GRANTZ



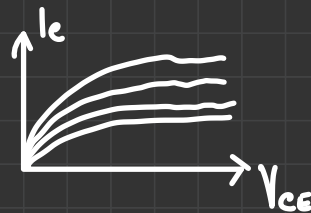
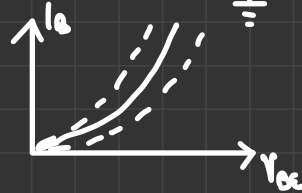
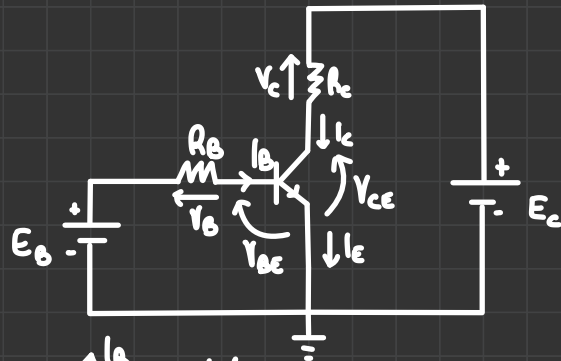
OR e AND



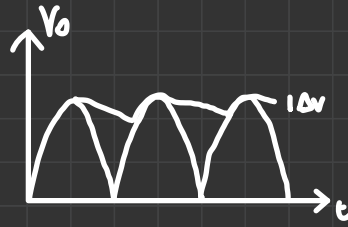
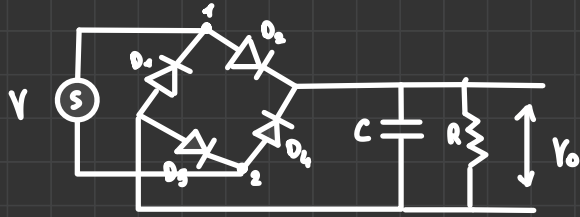
TRANSISTOR o BJT (BIPOLAR JUNCTION TRANSISTOR)



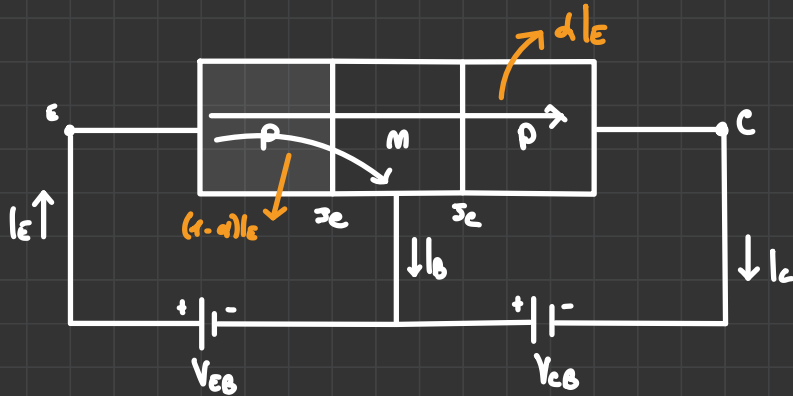
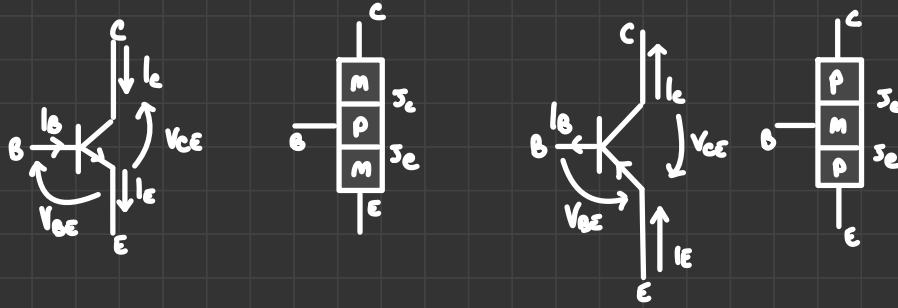
$$I_C = \alpha I_E \quad I_B = (1 - \alpha) I_E \quad I_E = I_B + I_C$$



PONTE DI GRAYZ



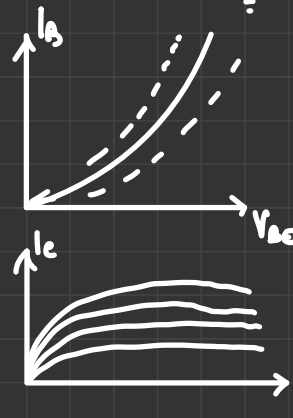
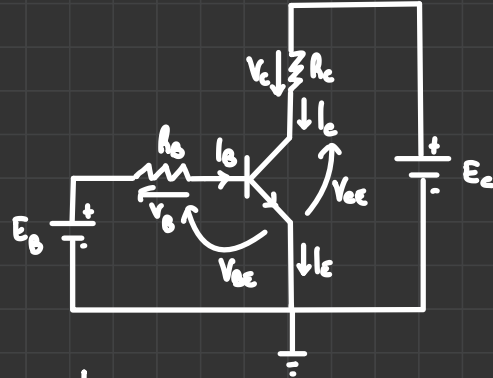
TRANSISTOR o BJT (BIPOLAR JUNCTION TRANSISTOR)



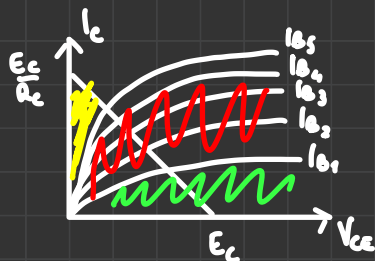
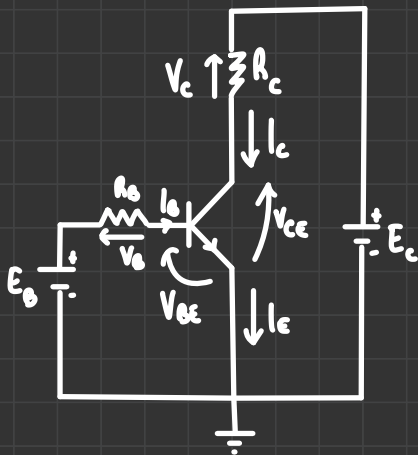
$$I_E = I_B + I_C$$

$$I_B = (1 - \alpha) I_C$$

$$I_E = \alpha I_C$$

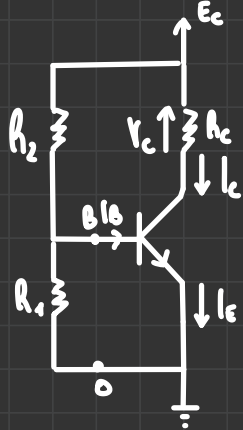


POLARIZZAZIONE

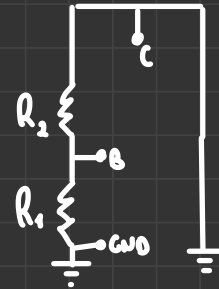
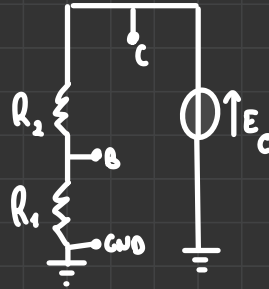


$$E_C = V_C + V_{CE} \rightarrow V_C = E_C - V_{CE} \rightarrow R_C I_C = E_C - V_{CE} \rightarrow I_C = \frac{E_C}{R_C} - \frac{V_{CE}}{R_C} \left\{ \begin{array}{l} V_{CE} = 0 \quad I_C = \frac{E_C}{R_C} \\ I_C = 0 \quad V_{CE} = E_C \end{array} \right.$$

POLARIZZAZIONE CON PARTITORE DI BASE



$$V_{B0} = E_B = E_c \frac{R_2}{R_1 + R_2}$$



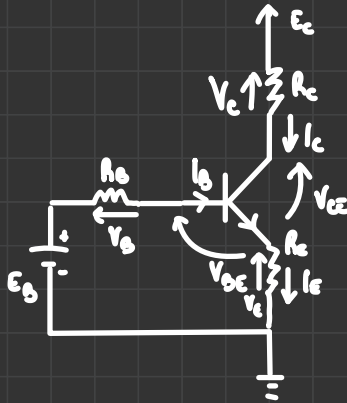
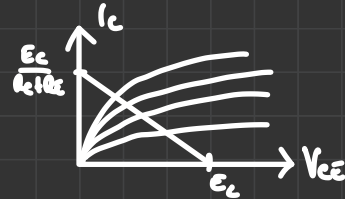
$$R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$I_c = \alpha I_e \approx I_e$$

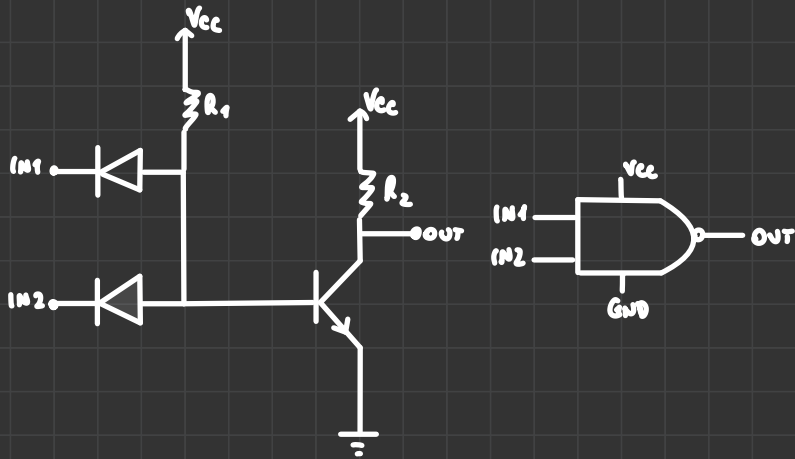
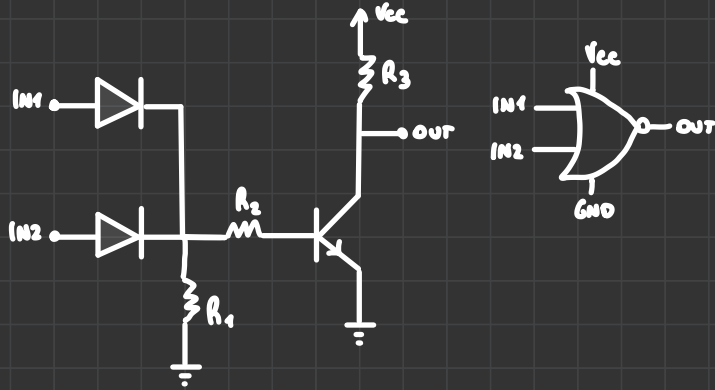
$$E_B = V_B + V_{BE} + V_E \rightarrow E_B = R_B I_B + V_{BE} + R_E I_E$$

$$E_C = V_C + V_{CE} + V_E \rightarrow E_C = R_C I_C + V_{CE} + R_E I_E \rightarrow R_E I_E + V_{CE} + R_C I_C$$

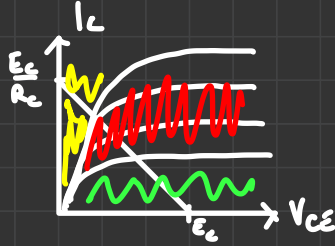
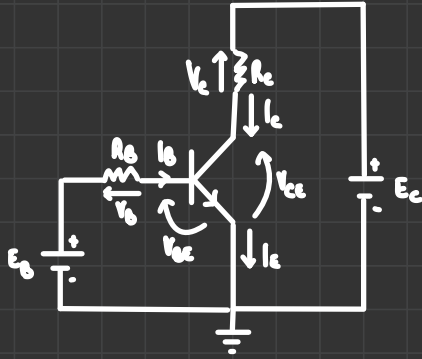
$$I_e = \frac{E_c}{R_c + R_E} - \frac{V_{ce}}{R_c + R_E}$$



PORTE LOGICHE NOR E NAND



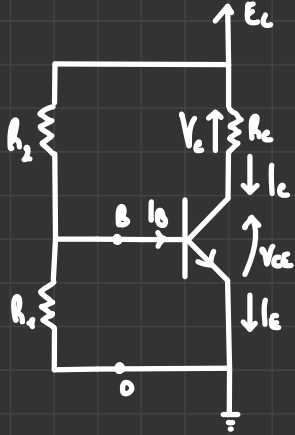
POLARIZZAZIONE



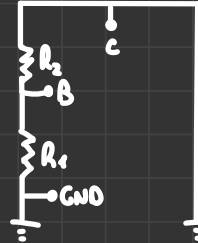
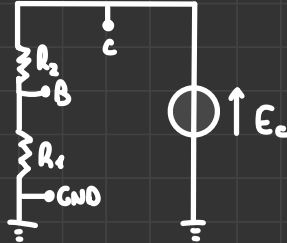
$$E_C = V_C + V_{CE} \rightarrow V_C = E_C - V_{CE} \rightarrow R_C I_C = E_C - V_{CE} \rightarrow I_C = \frac{E_C}{R_C} - \frac{V_{CE}}{R_C}$$

$$\begin{cases} V_{CE} = 0 & I_C = \frac{E_C}{R_C} \\ I_C = 0 & V_{CE} = E_C \end{cases}$$

POLARIZZAZIONE CON PARTITORE DI BASE

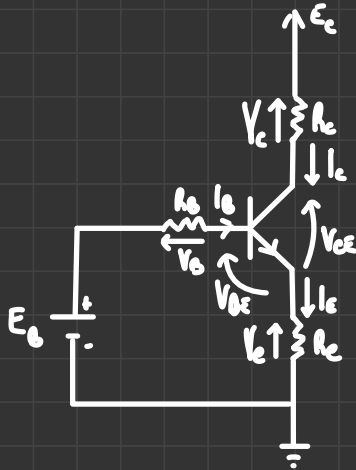


$$V_{B0} = E_B = E_c \frac{R_1}{R_1 + R_2}$$



$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$I_c = \alpha I_e \approx I_e$$



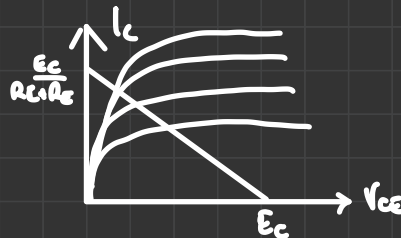
$$E_B = V_B + V_{BE} + V_E$$

$$E_B = R_B I_B + V_{BE} + R_E I_E$$

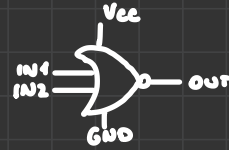
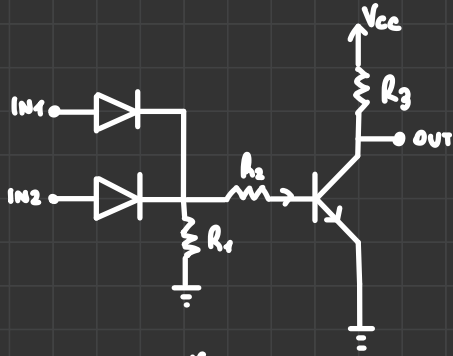
$$E_c = V_c + V_{CE} + V_E$$

$$E_c = R_C I_C + V_{CE} + R_E I_E \rightarrow E_c = R_C I_C + V_{CE} + R_E I_C$$

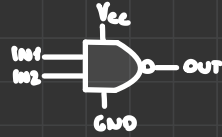
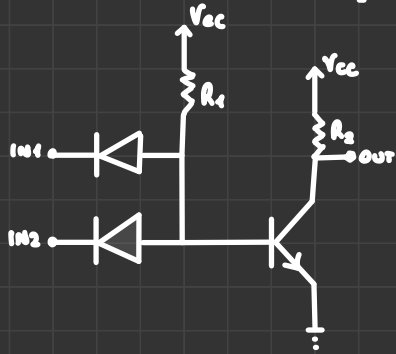
$$I_C = \frac{E_c}{R_C + R_E} - \frac{V_{CE}}{R_C + R_E}$$



PORTE LOGICHE NOR NAND

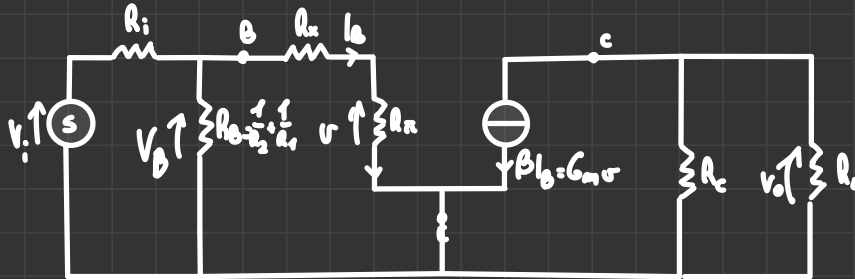
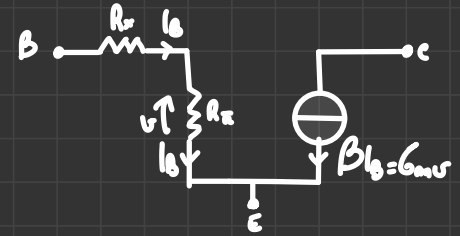
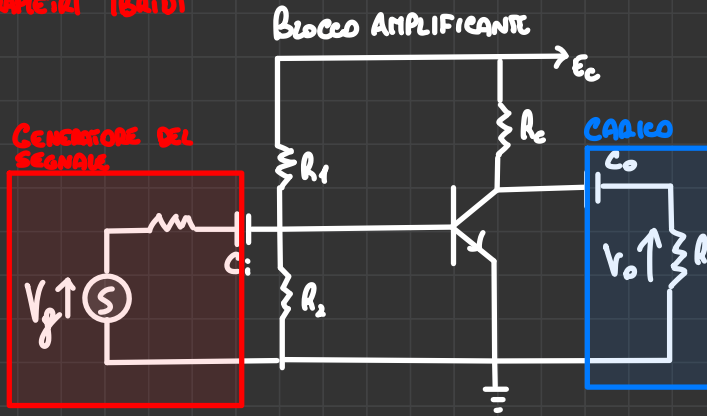


PORTA NOR



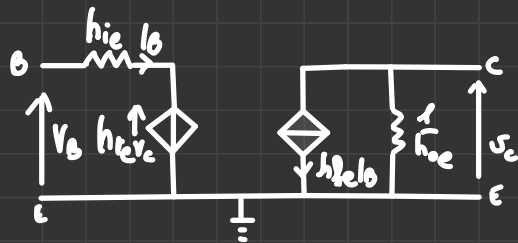
PORTA NAND

MODELLO A PARAMETRI IBIDI



$$A = \frac{v_o}{v_i}$$

Modelo A Parametros H



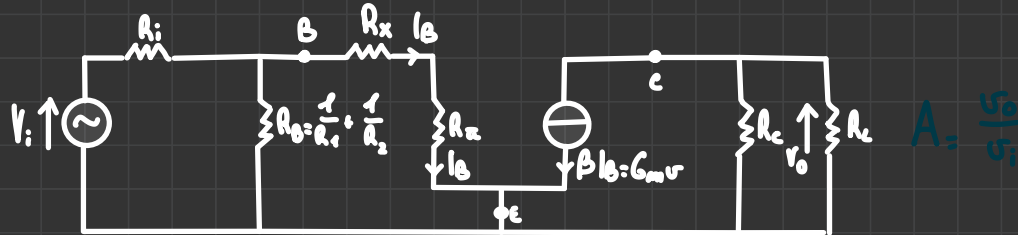
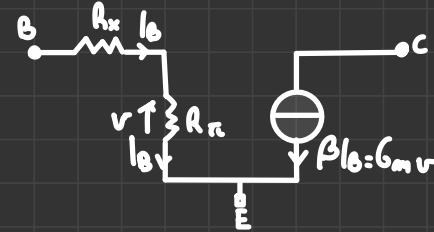
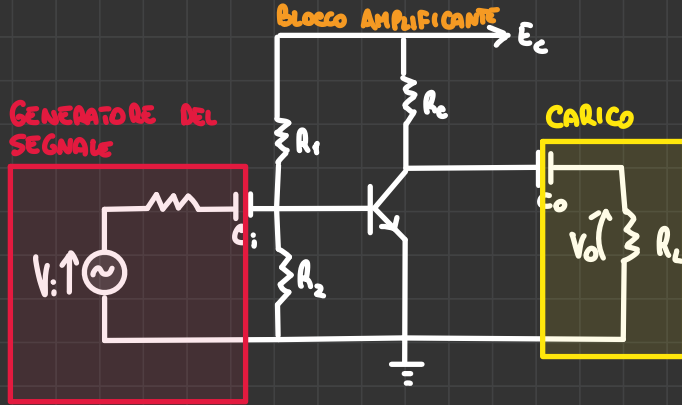
$$h_{ie} = R_x + R_z$$

$$h_{fe} = \beta$$

$$h_{re} = 0$$

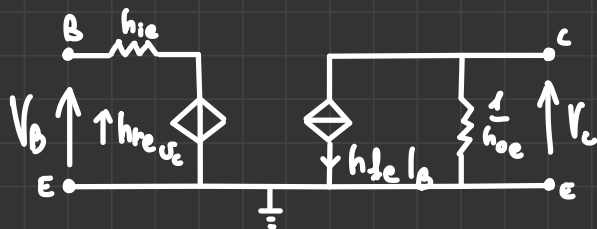
$$h_{oe} = 0$$

Modello A Parametri Ibridi / π Ibrido



$$A_v = \frac{v_o}{v_i}$$

Modello A Parametri H



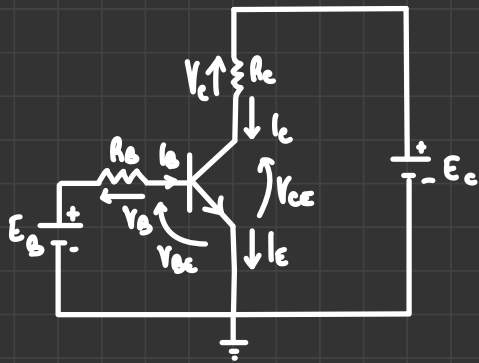
$$h_{ie} = R_x + r_\pi$$

$$h_{fe} = \beta$$

$$h_{re} = 0$$

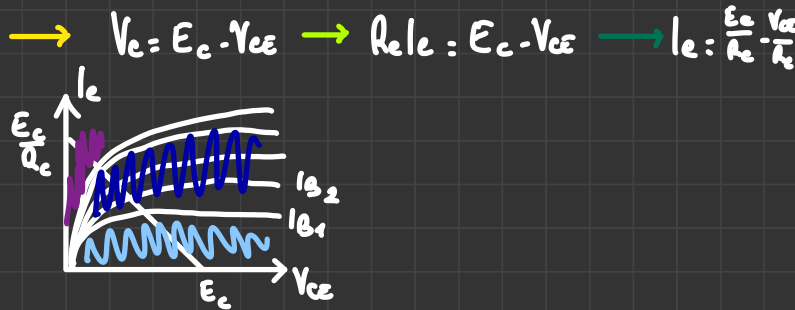
$$h_{oe} = 0$$

POLARIZZAZIONE

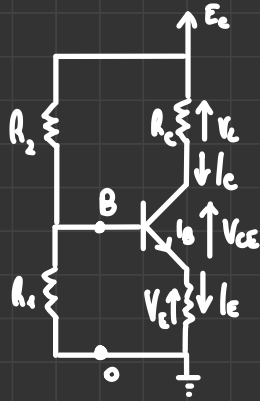


$$E_C = V_C + V_{CE}$$

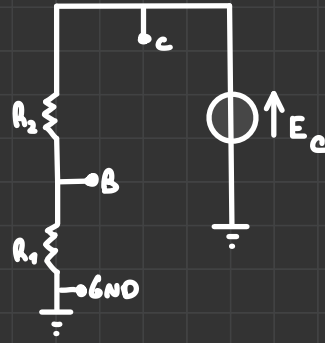
$$\left\{ \begin{array}{l} V_{CE} = 0 \\ I_E = 0 \end{array} \right. \quad \begin{array}{l} I_C = \frac{E_C}{R_C} \\ V_{CE} = E_C \end{array}$$



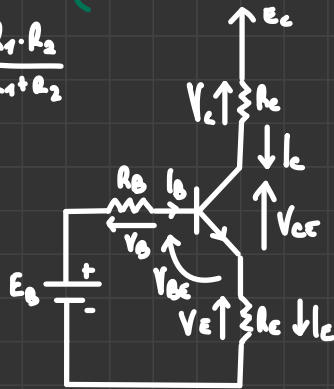
POLARIZZAZIONE CON PARTITORE DI BASE



$$V_{B0} = E_B = E_C \frac{R_1}{R_1 + R_2}$$



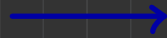
$$R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$



$$E_B = V_B + V_{CE} + V_E$$

$$E_C = V_C + V_{CE} + V_E$$

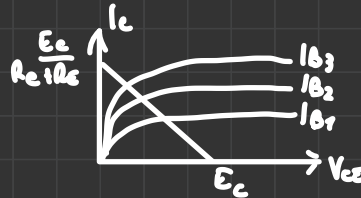
$$I_C = \alpha I_E \approx I_E$$



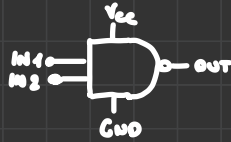
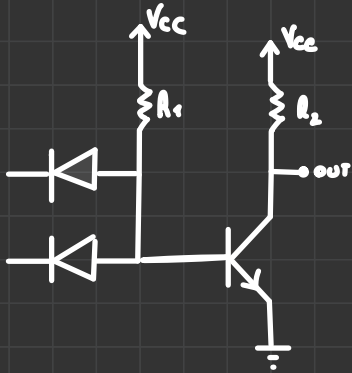
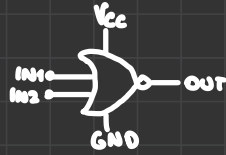
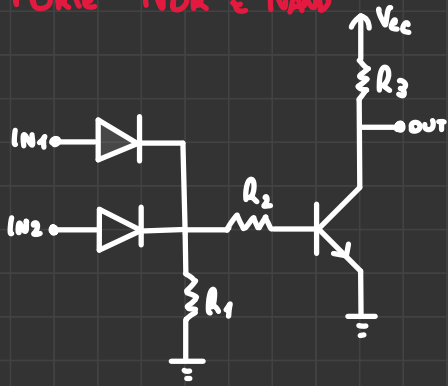
$$E_C = R_C I_C + V_{CE} + R_E I_E \rightarrow E_C = R_C I_C + V_{CE} + R_E I_C$$

$$I_C (R_C + R_E) = E_C - V_{CE}$$

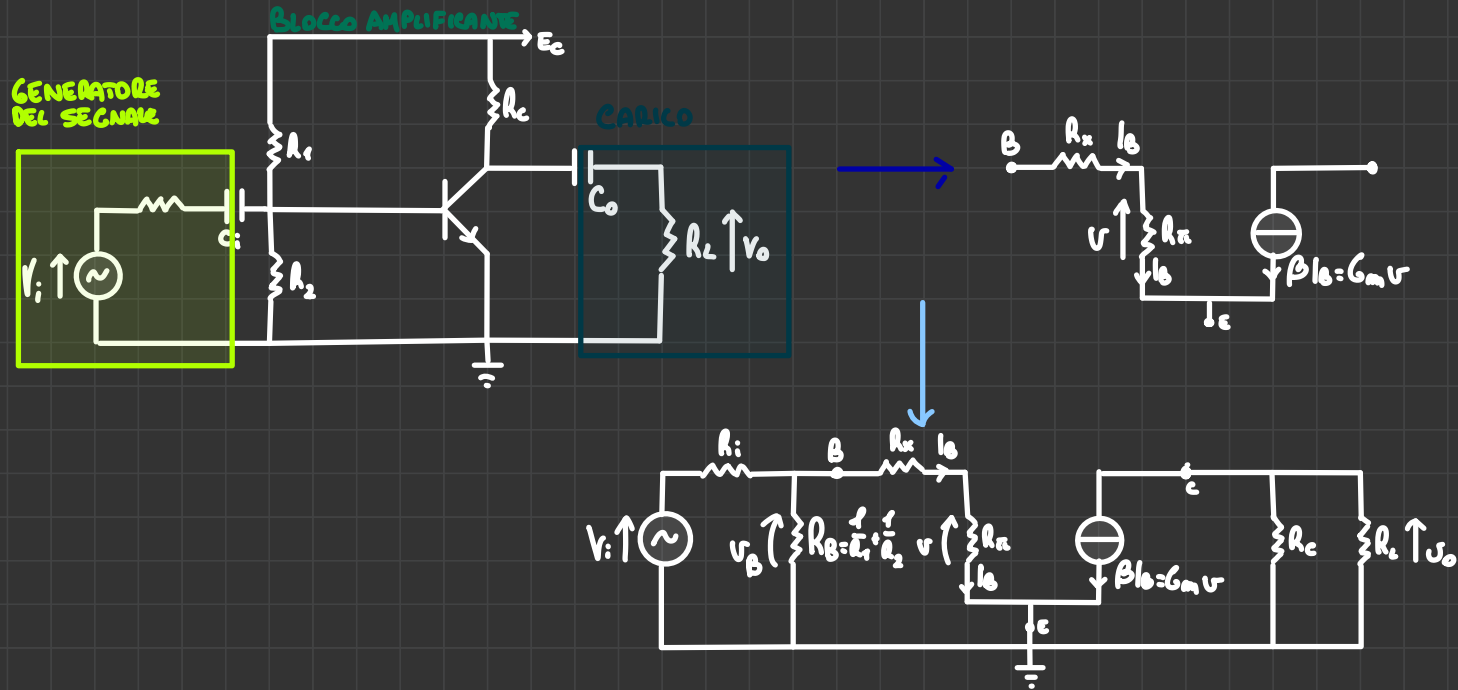
$$I_C = \frac{E_C}{R_C + R_E} - \frac{V_{CE}}{R_C + R_E}$$



Porte NOR e NAND

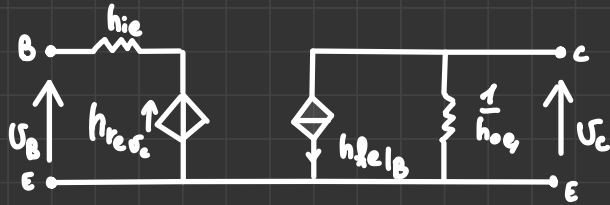


MODELLO A PARAMETRI IBRIDI/TE IBRIDI



$$A = \frac{V_o}{V_i}$$

Modello a parametri H



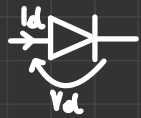
$$h_{ie} = R_x + R_n$$

$$h_{fe} = \beta$$

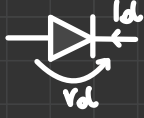
$$h_{re} = 0$$

$$h_{oe} = 0$$

DIODI



DIRETTAMENTE
POLARIZZATO



INVERSAMENTE
POLARIZZATO

$$\frac{\Delta V}{\Delta I} \rightarrow R_{\text{bassa}}$$

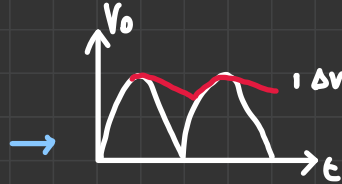
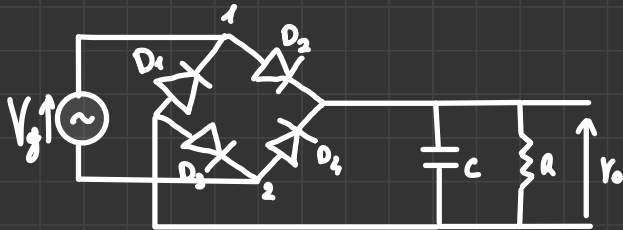
$$\frac{\Delta V}{\Delta I} \rightarrow R_{\text{alta}}$$

DIODO ZENER

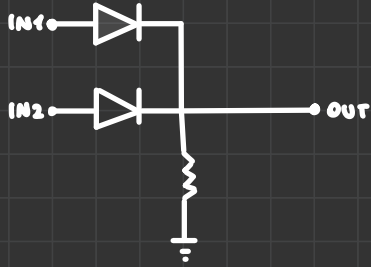
$$\frac{\Delta I}{\Delta V} \rightarrow \infty$$



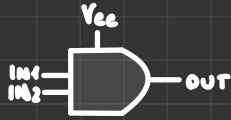
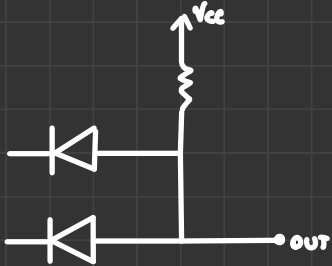
PONTE DI GRAYZ / RADDOLZATORE A DOPPIA SEMIONDA



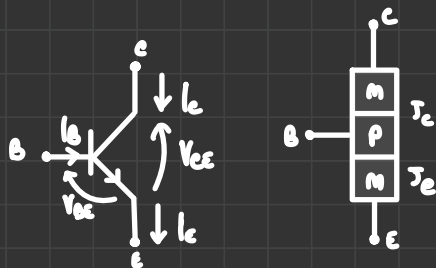
Porta OR



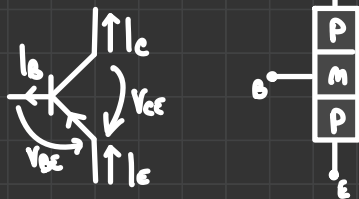
Porta AND



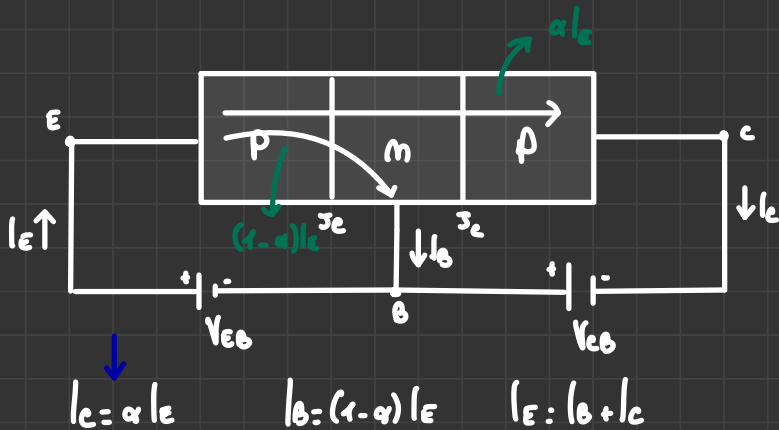
TRANSISTOR



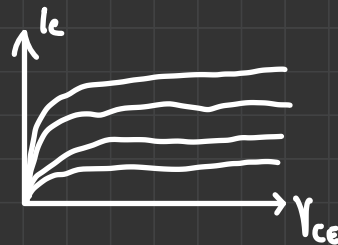
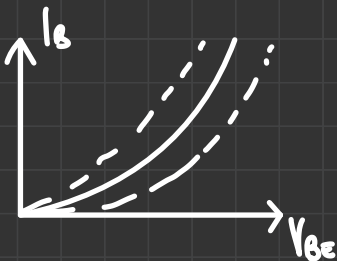
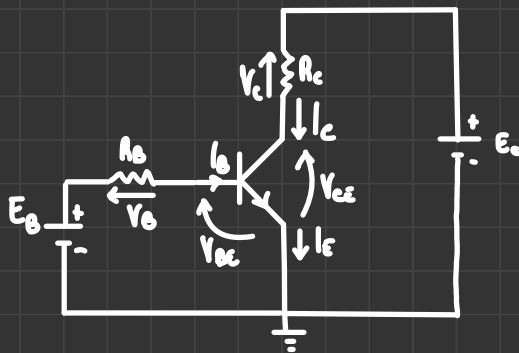
NPN



PNP

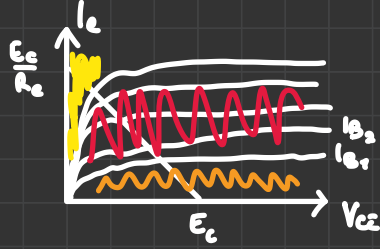


CARATTERISTICHE DI INGRESSO E DI USCITA

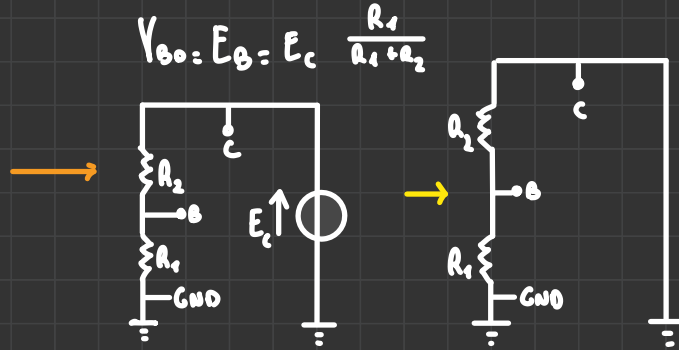


POLARIZZAZIONE

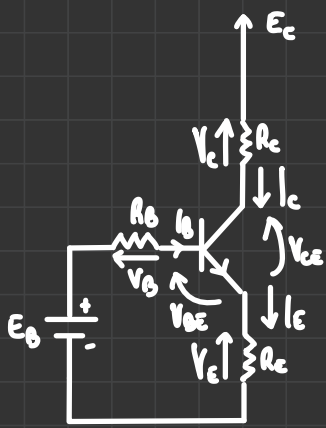
$$E_c = V_c + V_{ce} \rightarrow R_c I_c = E_c - V_{ce} \rightarrow I_c = \frac{E_c}{R_c} - \frac{V_{ce}}{R_c} \rightarrow \begin{cases} I_c = 0 & V_{ce} = E_c \\ V_{ce} = 0 & I_c = \frac{E_c}{R_c} \end{cases}$$



POLARIZZAZIONE CON PARTITORE DI BASE

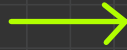


$$R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$



$$E_B = V_{BE} + V_{BE} + V_{BE}$$

$$E_C = V_C + V_{CE} + V_E$$

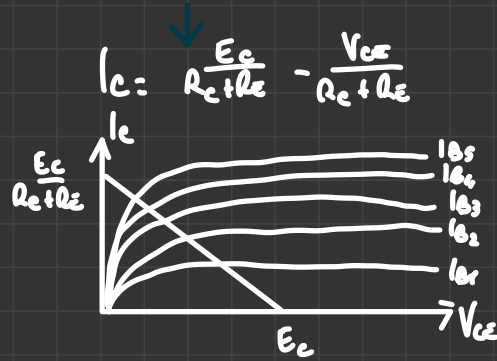


$$I_C = \alpha I_E \approx I_E$$

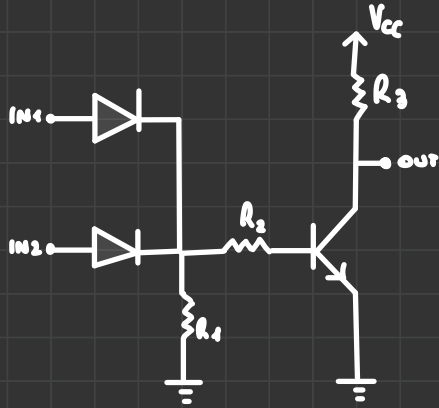
$$E_C = R_C I_C + V_{CE} + R_E I_E$$



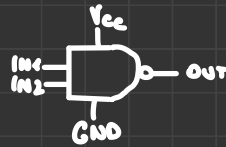
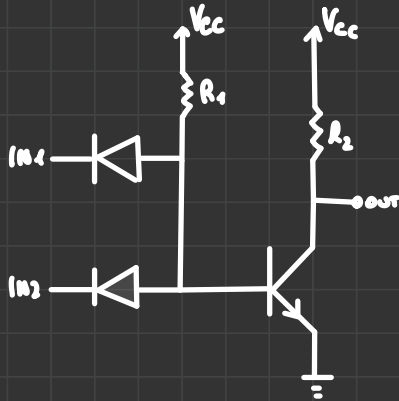
$$E_C = R_C I_C + V_{CE} + R_E I_C$$



PORTA NOR



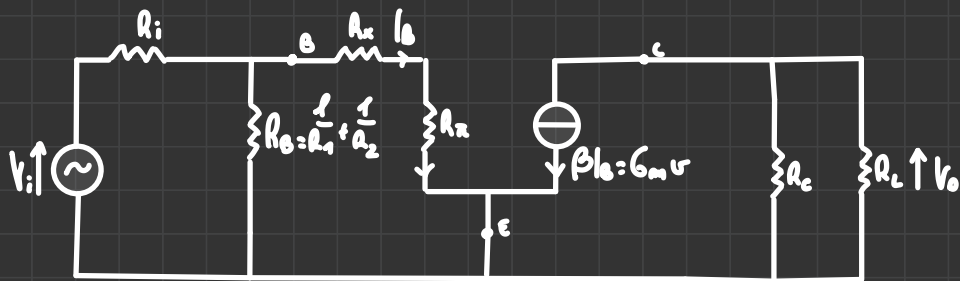
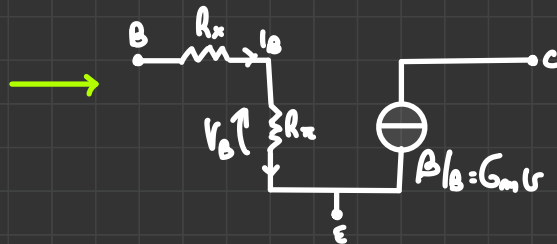
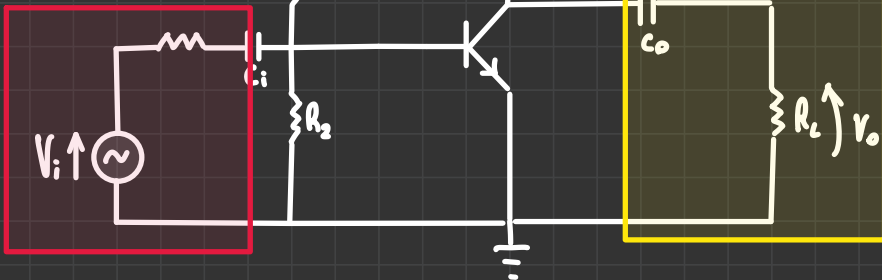
PORTA NAND



MODELLO A PARAMETRI IBAIDI / TC IBAIDI

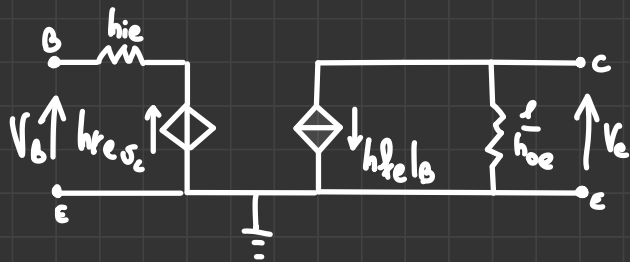
BLOCCO AMPLIFICANTE

GENERATORE DEL SEGNALE



$$A = \frac{V_o}{V_i}$$

Modello A Parametri H



$$h_{ie} = h_{\pi} + h_{\pi}$$

$$h_{fe} = \beta$$

$$h_{re} = 0$$

$$h_{oe} = 0$$

FILTRO PASSA-BASSO PASSIVO DEL 1° ORDINE

$$H(s) = \frac{U_c(s)}{E(s)} = \frac{\frac{1}{sC}}{\frac{1}{sC} + R} = \frac{1}{1 + sRC} = \frac{1}{RC(s + \frac{1}{RC})}$$

$$s = j\omega$$

$$H(j\omega) = \frac{1}{RC(j\omega + \frac{1}{RC})}$$

MODULO

$$|H(j\omega)| = \frac{1}{RC \sqrt{\omega^2 + (\frac{1}{RC})^2}}$$

$$\omega \rightarrow 0$$

$$\omega \rightarrow \infty$$

FASE

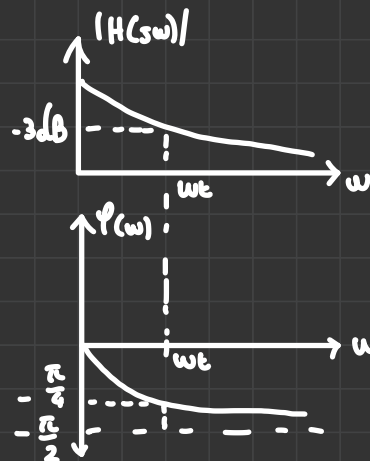
$$\varphi(\omega) = \text{ARCTAN}(-\omega RC)$$

PULSAZIONE DI TAGLIO

$$H(\omega_c) = \frac{1}{\sqrt{2}} = -3 \text{ dB}$$

$$\frac{1}{RC \sqrt{\omega^2 + (\frac{1}{RC})^2}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c^2 + (\frac{1}{RC})^2 = 2 \left(\frac{1}{RC}\right)^2 \rightarrow \omega_c = \frac{1}{RC} \rightarrow H(s) = \frac{\omega_c}{s + \omega_c}$$

$$\varphi(\omega_c) = \text{ARCTAN}\left(-\frac{1}{RC} \cdot RC\right) = \text{ARCTAN}(-1) = -\frac{\pi}{4}$$



FILTRO PASSA-BASSO PASSIVO DEL 1° ORDINE

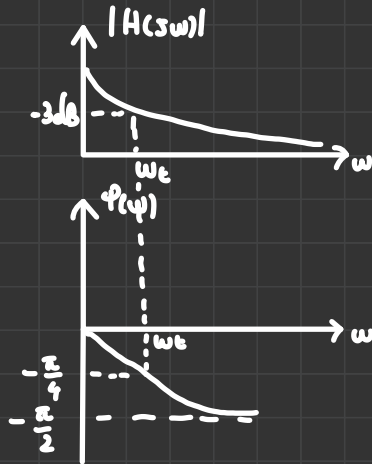
$$H(s) = \frac{V_o(s)}{E(s)} = \frac{\frac{1}{sC}}{\frac{1}{sC} + R} = \frac{1}{1 + R s C} = \frac{1}{RC \left(s + \frac{1}{RC}\right)}$$

$s = j\omega$

$$H(j\omega) = \frac{1}{RC(j\omega + \frac{1}{RC})}$$

$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 1$

$\omega \rightarrow \infty \quad H(j\omega) \rightarrow 0$



MODULO

$$|H(j\omega)| = \frac{1}{RC \sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}$$

$$\varphi(\omega) = \text{ARCTAN}(-\omega RC)$$

PULSAZIONE DI TAGLIO

$$H(j\omega) = \frac{1}{\sqrt{2}} = -3dB$$

$$\frac{1}{RC \sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}} = \frac{1}{\sqrt{2}} \rightarrow \omega^2 + \left(\frac{1}{RC}\right)^2 = 2 \left(\frac{1}{RC}\right)^2$$
$$\omega_c = \frac{1}{RC}$$

$$H(s) = \frac{\omega_c}{(s + \omega_c)}$$

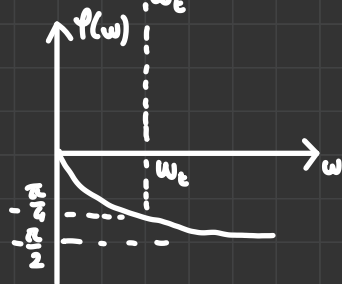
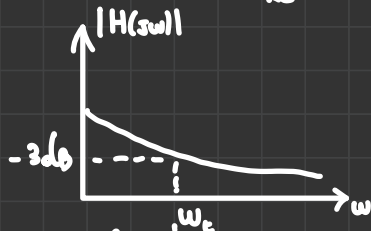
$$\varphi(\omega_c) = \text{ARCTAN}\left(-\frac{1}{RC} \cdot RC\right) = \text{ARCTAN}(-1) = -\frac{\pi}{4}$$

FILTRO PASSA-BASSO PASSIVO 1° ORDINE

$$H(s) = \frac{V_o(s)}{E(s)} = \frac{\frac{1}{sC}}{\frac{1}{sC} + R} = \frac{1}{1 + sRC} = \frac{1}{RC(s + \frac{1}{RC})}$$

$$s = j\omega$$

$$H(j\omega) = \frac{1}{RC(j\omega + \frac{1}{RC})}$$



MODULO

$$|H(j\omega)| = \frac{1}{RC \sqrt{\omega^2 + (\frac{1}{RC})^2}}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 1$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 0$$

FASE

$$\varphi(\omega) = \text{ARCTAN}(-\omega RC)$$

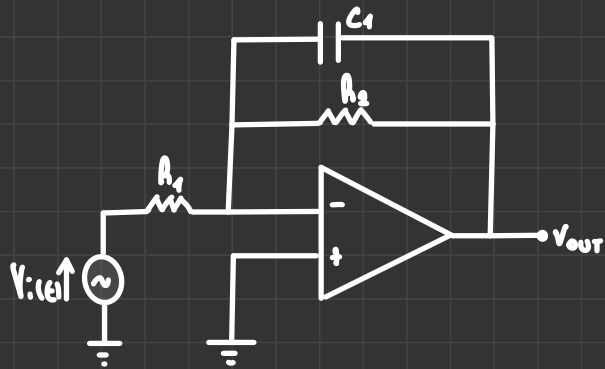
PULSABEQUE DI TAGLIO

$$H(j\omega_c) = \frac{1}{\sqrt{2}} = -3\text{dB}$$

$$\frac{1}{RC \sqrt{\omega_c^2 + (\frac{1}{RC})^2}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c^2 + (\frac{1}{RC})^2 = 2(\frac{1}{RC})^2 \rightarrow \omega_c = \frac{1}{RC} \rightarrow H(s) = \frac{\omega_c}{s + \omega_c}$$

$$\varphi(\omega_c) = \text{ARCTAN}\left(-\frac{1}{RC} RC\right) = \text{ARCTAN}(-1) = -\frac{\pi}{4}$$

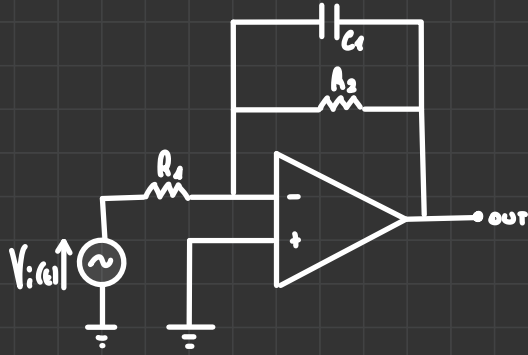
FILTRO PASSA-BASSO ATTIVO DEL 1° ORDINE



$$H(s) = - \frac{G_1}{sC_1 + G_2} = - \frac{\frac{G_1}{C_1}}{s + \frac{G_2}{C_1}}$$
$$= - \frac{\omega_c}{s + \omega_c}$$

$$\omega_c = \frac{G_1}{C_1} \quad G_1 = G_2$$

FILTRO PASSA BASSO ATTIVO DEL 1° ORDINE



$$H(s) = -\frac{G_1}{sC_1 + G_2} = -\frac{\frac{G_1}{C_1}}{s + \frac{G_2}{C_1}} = -\frac{\omega_c}{s + \omega_c} \quad \omega_c = \frac{G_1}{C_1} \quad G_2 = G_1$$

FILTRO PASSA-ALTO PASSIVO DEL 1° ORDINE

$$H(s) = 1 - H_{LP}(s) = 1 - \frac{\omega_c}{s + \omega_c} = \frac{s}{s + \omega_c}$$

$s = j\omega$

$$\omega_c = \frac{1}{RC}$$

$$H(j\omega) = \frac{j\omega}{j\omega + \omega_c}$$

MODULO

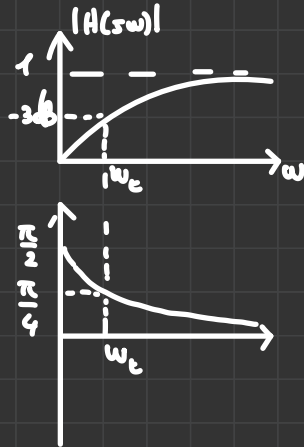
$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + \omega_c^2}}$$

FASE

$$\varphi(\omega) = \text{ARCTAN}\left(\frac{\omega}{\omega_c}\right)$$

$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 1$
 $\omega \rightarrow \infty \quad H(j\omega) \rightarrow 0$

$$\varphi(\omega_c) = \text{ARCTAN}(1) = \frac{\pi}{4}$$



FILTRO PASSA-ALTO PASSIVO DEL 1° ORDINE

$$H(s) = 1 - H_{LP}(s) = 1 - \frac{\omega_c}{s + \omega_c} = \frac{s}{s + \omega_c}$$

$$\omega_c = \frac{1}{RC}$$

$$s = j\omega$$

MODULO

FASE

$$H(j\omega) = \frac{j\omega}{j\omega + \omega_c}$$

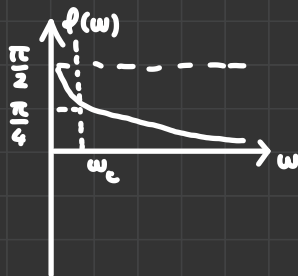
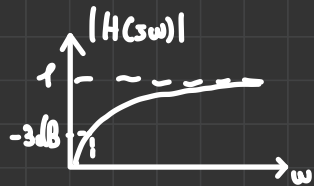
$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + \omega_c^2}}$$

$$\varphi(\omega) = \text{ARCTAN}\left(\frac{\omega}{\omega_c}\right)$$

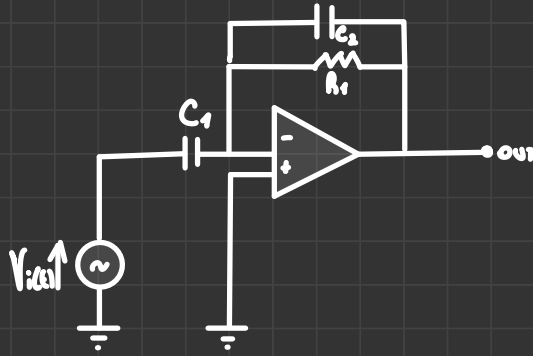
$$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 0$$

$$\omega \rightarrow \infty \quad H(j\omega) \rightarrow 1$$

$$\varphi(\omega_c) = \text{ARCTAN}(1) = \frac{\pi}{4}$$



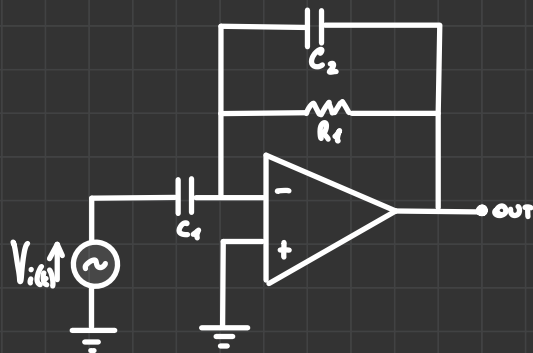
FILTRO PASSA-ALTO ATTIVO DEL 1° ORDINE



$$H(s) = \frac{C_1}{C_2} \frac{s}{s + \frac{G_1}{C_2}} = - \frac{s}{s + \omega_c}$$

$$\omega_c = \frac{G_1}{C_2} \quad \text{e} \quad C_1 = C_2$$

FILTRO PASSA-ALTO ATIVO DEL 1 ORDINE



$$H(s) = -\frac{C_1}{C_2} \frac{s}{s + \frac{G_1}{C_2}} = -\frac{s}{s + \omega_c}$$

$$\omega_c = \frac{G_1}{C_2} \quad C_1 = C_2$$

FILTRO PASSA-ALTO PASSIVO DEL 1° ORDINE

$$H(s) = 1 - H_{LP}(s) = 1 - \frac{\omega_c}{s + \omega_c} = \frac{s}{s + \omega_c} \quad \omega_c = \frac{1}{RC}$$

$$s = j\omega$$

$$H(j\omega) = \frac{j\omega}{j\omega + \omega_c}$$

MODULO

$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + \omega_c^2}}$$

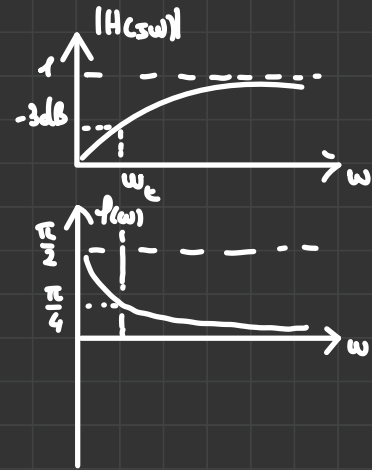
$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 0$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 1$$

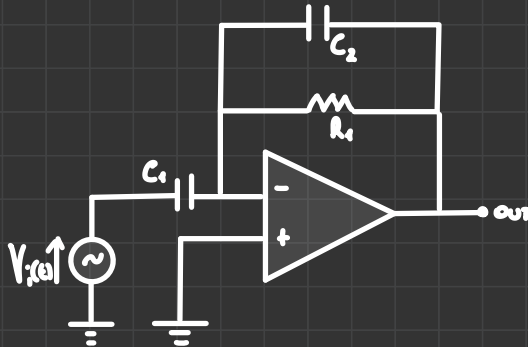
FASE

$$\varphi(\omega) = \text{ARCTAN}\left(\frac{\omega}{\omega_c}\right)$$

$$\varphi(\omega_c) = \text{ARCTAN}(1) = \frac{\pi}{4}$$



FILTRO PASSA ALTO ATTIVO DEL PRIMO ORDINE



$$H(s) = \frac{C_1}{C_2} \frac{s}{s + \frac{G_1}{C_2}} = - \frac{s}{s + \omega_c}$$

$$\omega_c = \frac{G_1}{C_2} \quad \text{E} \quad C_1 = C_2$$

FILTRO PASSA BANDA PASSIVO DEL II ORDINE

$$H(s) = \frac{R}{R + \frac{1}{sC} + sL} = \frac{sCR}{sCR + 1 + s^2LC} = \frac{s \cancel{R}}{\cancel{L} \left(\frac{R}{L}s + \frac{1}{LC} + s^2 \right)} \quad B = \frac{R}{L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\rightarrow = \frac{Bs}{(Bs + \omega_0^2 + s^2)}$$

$$s = j\omega$$

$$H(j\omega) = \frac{Bj\omega}{(-\omega^2 + Bs + \omega_0^2)}$$

MODULO

$$|H(j\omega)| = \frac{\omega B}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\omega B)^2}}$$

$$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 0$$

$$\omega = \omega_0 \quad H(j\omega)$$

$$\omega \rightarrow \infty \quad H(j\omega) \rightarrow 0$$

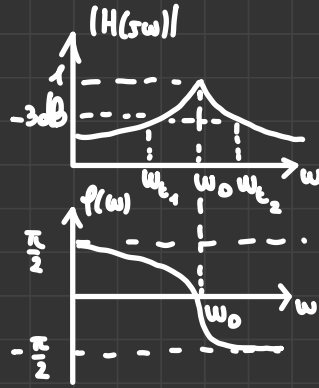
FASE

$$\varphi(\omega) = \arctan\left(\frac{\omega_0^2 - \omega^2}{\omega B}\right)$$

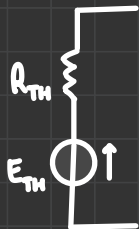
$$\varphi(0) = \frac{\pi}{2}$$

$$\varphi(\omega_0) = 0$$

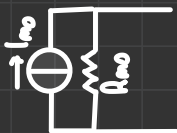
$$\varphi(\infty) = -\frac{\pi}{2}$$



TEOREMA DI THEVENIN E NORTON



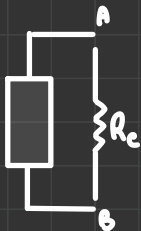
FORMA THEVENIN



FORMA NORTON

$$\longrightarrow I_{NO} = \frac{E_{TH}}{R_{TH}} \quad R_{TH} = R_{NO}$$

TEOREMA DEL MASSIMO TRASFERIMENTO DI POTENZA



$$\begin{cases} P_{max} = \frac{E_{TH}^2}{4R_{TH}} \\ R_L = R_{eq} = R_{TH} \end{cases}$$

$$V_{AB} = E_{TH} \frac{R_L}{R_L + R_{TH}}$$

$$i = \frac{V}{R_L} = \frac{E_{TH}}{R_L + R_{TH}}$$

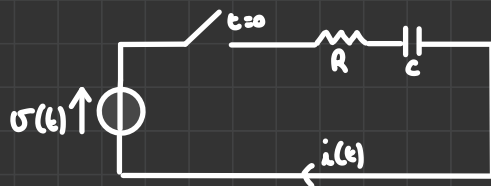
$$P = R_L \cdot i^2 = R_L \frac{E_{TH}^2}{(R_L + R_{TH})^2} = R_L \frac{E_{TH}^2}{R_L^2 + R_{TH}^2 + 2R_L R_{TH}} = \frac{E_{TH}^2}{R_L + \frac{R_{TH}^2}{R_L} + 2R_{TH}}$$

$$\frac{d}{dR_L} \left(R_L + \frac{R_{TH}^2}{R_L} + 2R_{TH} \right) = -\frac{R_{TH}^2}{R_L^2} + 1 = 0$$

$$R_L = R_{TH}$$

$$P_{max} = R_{TH} \frac{E_{TH}^2}{(2R_{TH})^2} = R_{TH} \frac{E_{TH}^2}{4R_{TH}^2} = \frac{E_{TH}^2}{4R_{TH}}$$

CARICA E SCARICA DEL CONDENSATORE



$$i(t) \begin{cases} 0 & t \leq 0^- \\ i(t) & t > 0^+ \end{cases}$$

$$e(t) = E u_{-1}(t)$$

$$v(t) = v_R(t) + v_C(t) \quad t \geq 0^+$$

$$E = Ri + v_C$$

$$E = RC \frac{dv_C}{dt} + v_C$$

$$v_C(t) = v_{Cg}(t) + v_{Cp}(t)$$

$$RC \frac{dv_C}{dt} + v_C = 0 \rightarrow RC\alpha + 1 = 0$$

$$\alpha = -\frac{1}{RC} \rightarrow v_{Cg}(t) = k_1 e^{-\frac{1}{RC}t}$$

$$v_{Cp}(t) = k_2$$

$$\begin{cases} v_C(0^+) = k_1 + k_2 \\ \lim_{t \rightarrow \infty} v_C(t) = E = k_2 \end{cases}$$

$$\begin{cases} k_1 = v_C(0^+) - E \\ k_2 = E \end{cases}$$

$$\begin{cases} v_C(t) = (v_C(0^+) - E) e^{-\frac{1}{RC}t} + E & t \geq 0^+ \\ i(t) = -\frac{1}{R} (v_C(0^+) - E) e^{-\frac{1}{RC}t} \end{cases}$$

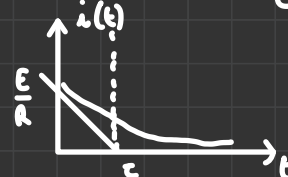
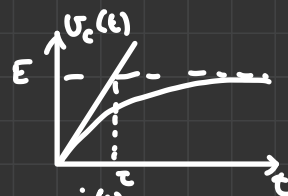
$$|v_C(0^+)| < |E| \quad \text{CARICA} \quad |v_C(0^+)| > |E| \quad \text{SCARICA}$$

$$v_C(0^+) = 0$$

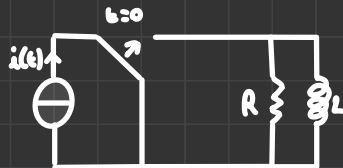
$$v_C(t) = E(1 - e^{-\frac{1}{RC}t}) u_{-1}(t)$$

$$i(t) = \frac{E}{R} e^{-\frac{1}{RC}t}$$

$$RC = \tau$$



CARICA E SCARICA DELL'INDUTTORE



$$i(t) = i_{Lg}(t) + i_{Lp}(t)$$

$$i(t) - i_{Lg}(t) - i_{Lp}(t) = 0 \quad t \geq 0^+$$

$$I = GL \frac{di_L}{dt} + i_L$$

$$i_L(t) = i_{Lg}(t) + i_{Lp}(t)$$

$$GL \frac{di_L(t)}{dt} + i_L(t) = 0$$

$$GL \frac{d}{dt} + 1 = 0 \quad \alpha = -\frac{1}{GL}$$

$$i_{Lg}(t) = k_1 e^{-\frac{1}{GL}t} \quad i_{Lp}(t) = k_2$$

$$i_L(t) = k_1 e^{-\frac{1}{GL}t} + k_2$$

$$\begin{cases} i_L(0^+) = k_1 + k_2 \\ \lim_{t \rightarrow \infty} i_L(t) = I = k_2 \end{cases}$$

$$\begin{cases} k_1 = (i_L(0^+) - I) \\ k_2 = I \end{cases}$$

$$\begin{cases} i_L(t) = (i_L(0^+) - I) e^{-\frac{1}{GL}t} + I & t \geq 0^+ \\ v(t) = \frac{1}{C} (i_L(0^+) - I) e^{-\frac{1}{GL}t} & t \geq 0^+ \end{cases}$$

$$|i_L(0^+)| < I \quad \text{CARICA}$$

$$|i_L(0^+)| > I \quad \text{SCARICA}$$

$$i_L(0^+) = 0 \quad GL = \tau$$

$$\begin{aligned} i_L(t) &= I(1 - e^{-\frac{1}{GL}t}) \\ v(t) &= \frac{1}{C}(e^{-\frac{1}{GL}t}) \end{aligned}$$



CIRCUITI IN REGIME SINUSOIDALE E FASORI

$$v(t) = V_m \sin(\omega t + \gamma) \longrightarrow v(t) = \frac{V_m e^{j(\omega t + \gamma)} - V_m e^{-j(\omega t + \gamma)}}{2}$$

$$i(t) = I_m \cos(\omega t + \beta) \longrightarrow i(t) = \frac{I_m e^{j(\omega t + \beta)} - I_m e^{-j(\omega t + \beta)}}{2}$$

$$\bar{V} = \frac{V_m e^{j\gamma}}{2j} \quad \bar{I} = \frac{I_m e^{j\beta}}{2j}$$

$$v(t) = \bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}$$

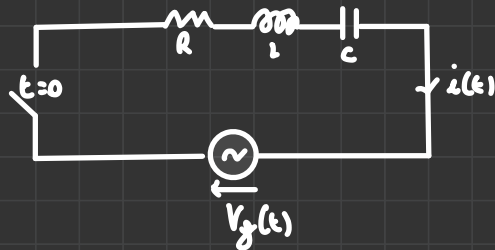
$$i(t) = \bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}$$

$$\bar{V} = V_m e^{j\gamma} \quad \bar{I} = I_m e^{j\beta}$$

$$v(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j}$$

$$i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j}$$

FASORI NEI CIRCUITI RLC SERIE CON FORZANTI SINUSOIDALI



$$V_g(t) = V_m \sin(\omega t + \gamma)$$

$$\bar{V}_g = R\bar{I} + j\omega L + \frac{1}{j\omega C} = \underline{R + j(\omega L - \frac{1}{\omega C})} \bar{I}$$

$$\begin{cases} V_g(t) = R i(t) + L \frac{di(t)}{dt} + V_C(t) \\ i(t) = C \frac{dV_C(t)}{dt} \end{cases}$$

$$\bar{V} = Z \bar{I}$$

$$V_{g_p}(t) = V_m \sin(\omega t + \theta) \quad V_{g_p}(0) = V_m \sin(\theta)$$

$$i(t) = I_m \sin(\omega t + \beta)$$

$$V_{g_p}(t) = \frac{1}{C} \int_0^t i(t) dt + V_{g_p}(0)$$

$$i_p(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j} \quad V(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j}$$

STUDIO L'IMPIEDENZA AL VARIARE DI ω NEI CIRCUITI RLC SERIE

$$Z(\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$|Z(\omega)| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\varphi_z(\omega) = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$



$\omega < \omega_0$ OHMICO-CAPACITIVO

$R=0$ PURAM. CAPACITIVO

$\omega > \omega_0$ OHMICO-INDUTTIVO

$R=0$ PURAM. INDUTTIVO

$$\omega \rightarrow \infty \quad \omega L \gg \frac{1}{\omega C} \quad \varphi_z(\omega) \rightarrow \frac{\pi}{2}$$

$$\omega = \omega_0 \quad \varphi_z(\omega) = 0$$

$$\omega \rightarrow 0 \quad \frac{1}{\omega C} \gg \omega L \quad \varphi_z(\omega) \rightarrow -\frac{\pi}{2}$$

POTENZA IN REGIME PERMANENTE SINUSOIDALE

$$v(t) = V_m \sin(\omega t + \alpha) \longrightarrow v(t) = \frac{\bar{v} e^{j\omega t} - \bar{v}^* e^{-j\omega t}}{2j}$$

$$i(t) = I_m \sin(\omega t + \beta) \longrightarrow i(t) = \frac{\bar{i} e^{j\omega t} - \bar{i}^* e^{-j\omega t}}{2j}$$

$$p = v(t) i(t) = \left(\frac{\bar{v} e^{j\omega t} - \bar{v}^* e^{-j\omega t}}{2j} \right) \left(\frac{\bar{i} e^{j\omega t} - \bar{i}^* e^{-j\omega t}}{2j} \right) = -\frac{1}{4} \left(\bar{v} \bar{i} e^{2j\omega t} - \bar{v} \bar{i}^* - \bar{v}^* \bar{i} + (\bar{v} \bar{i})^* e^{-2j\omega t} \right)$$

$$= \underbrace{-\frac{1}{2} \frac{\bar{v} \bar{i} e^{2j\omega t} + (\bar{v} \bar{i})^* e^{-2j\omega t}}{2}}_{\text{VAR. NEL TEMPO}} + \underbrace{\frac{1}{2} \frac{\bar{v} \bar{i}^* + \bar{v}^* \bar{i}}{2}}_{\text{COSTANTE}}$$

$$\bar{v} \bar{i} = V_m I_m e^{j(\alpha + \beta)} \quad \bar{v} \bar{i}^* = V_m I_m e^{j(\alpha - \beta)}$$

$$(\bar{v} \bar{i})^* = V_m I_m e^{-j(\alpha + \beta)} \quad \bar{v}^* \bar{i} = V_m I_m e^{-j(\alpha - \beta)}$$

$$\alpha - \beta = \varphi$$

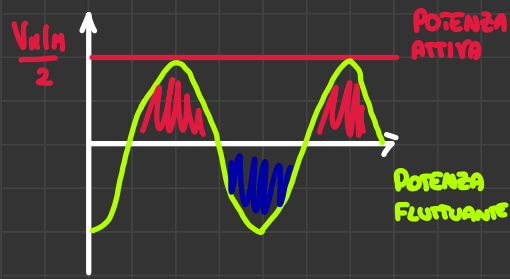
$$\alpha + \beta = 2\alpha - \varphi$$

$$= \frac{1}{2} V_m I_m \frac{e^{j(2\alpha + 2\alpha - \varphi)} + e^{-j(2\alpha + 2\alpha - \varphi)}}{2} + \frac{1}{2} V_m I_m \frac{e^{j\varphi} - e^{-j\varphi}}{2} = \frac{1}{2} V_m I_m \cos(2\omega t + 2\alpha - \varphi) + \frac{1}{2} V_m I_m \cos(\varphi)$$

POTENZA FLUTTUANTE POTENZA ATTIVA

$$\begin{cases} V_{\text{EFF}} = \frac{V_m}{\sqrt{2}} \rightarrow V_m = V_{\text{EFF}} \sqrt{2} \\ I_{\text{EFF}} = \frac{I_m}{\sqrt{2}} \rightarrow I_m = I_{\text{EFF}} \sqrt{2} \end{cases}$$

$$= V_{\text{EFF}} I_{\text{EFF}} \cos(\varphi) - V_{\text{EFF}} I_{\text{EFF}} \cos(2\omega t + 2\alpha - \varphi)$$



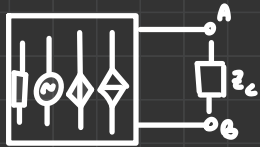
$$P_c = \nabla T^*$$

$$\dot{P}_c = P + jQ$$

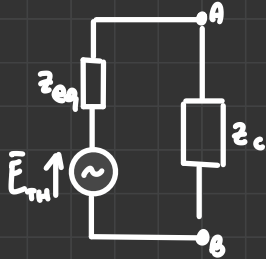
$$P = |P_c| \cos(\varphi)$$

$$Q = |P_c| \sin(\varphi)$$

TEOREMA DEL MASSIMO TRASFERIMENTO DI POTENZA PER RETI ADINAMICHE



$$Z_c = Z_{eq}^*$$



$$\bar{I} = \frac{E_{TH}}{Z_{eq} + Z_c} = \frac{E_{TH}}{(R_c + jX_c) + (R_{eq} + jX_{eq})}$$

$$P_c = R \bar{I}^2 + jX \bar{I}^2 \quad P = R_c \bar{I}^2 = R_c \frac{E_{TH}^2}{(R_c + R_{eq})^2 + (X_{eq} + X_c)^2}$$

$$X_c = -X_{eq} \quad X_{TOT} = 0 \quad Z_c = Z_{eq}^*$$

FILTRO PASSA-BASSO PASSIVO DEL 1° ORDINE

$$H(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{sRC + 1} = \frac{1}{RC(s + \frac{1}{RC})}$$

$$s = j\omega$$

$$H(j\omega) = \frac{1}{RC(j\omega + \frac{1}{RC})}$$

$$|H(j\omega)| = \frac{1}{RC \sqrt{\omega^2 + (\frac{1}{RC})^2}}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 1$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 0$$

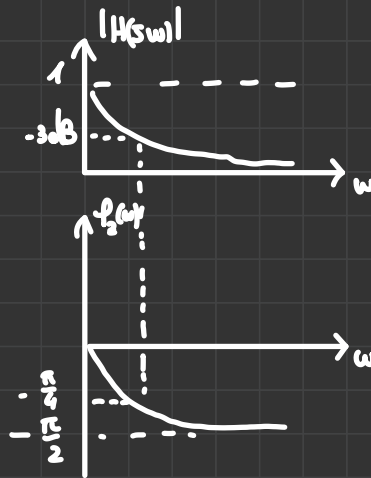
$$\varphi_2(\omega) = \text{ARCTAN}(-\omega RC)$$

$$|H(j\omega)| = \frac{1}{\sqrt{2}} = -3 \text{ dB}$$

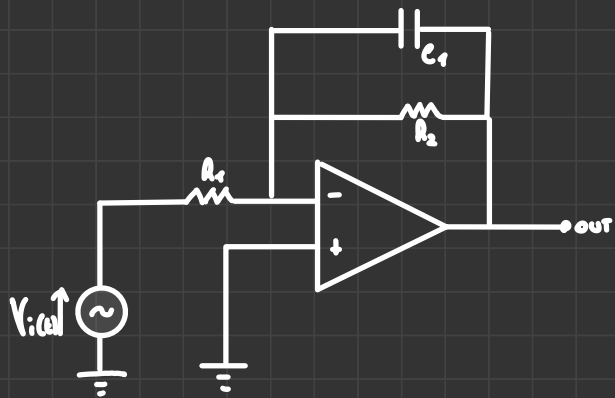
$$\frac{1}{RC \sqrt{\omega_c^2 + (\frac{1}{RC})^2}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c^2 + (\frac{1}{RC})^2 = 2 \left(\frac{1}{RC}\right)^2$$

$$\omega_c = \frac{1}{RC}$$

$$H(s) = \frac{\omega_c}{s + \omega_c} \quad \varphi_2(\omega_c) = \text{ARCTAN}\left(-\frac{1}{RC} RC\right) = \text{ARCTAN}(-1) = -\frac{\pi}{4}$$



FILTRO PASSA-BASSO ATTIVO DEL 1° ORDINE



$$H(s) = - \frac{G_1}{sC_1 + G_2} = - \frac{\frac{G_1}{C_1}}{s + \frac{G_2}{C_1}} = - \frac{\omega_c}{s + \omega_c}$$

$$\omega_c = \frac{G_1}{C_1} \quad \text{e} \quad G_1 = G_2$$

FILTRO PASSA-ALTO PASSIVO DEL 1° ORDINE

$$H(s) = 1 - H_{LP}(s) = 1 - \frac{\omega_c}{s + \omega_c} = \frac{s}{s + \omega_c}$$

$$s = j\omega$$

$$H(j\omega) = \frac{j\omega}{j\omega + \omega_c}$$

MODULO

$$|H(j\omega)| = \frac{\omega}{\sqrt{-\omega^2 + \omega_c^2}}$$

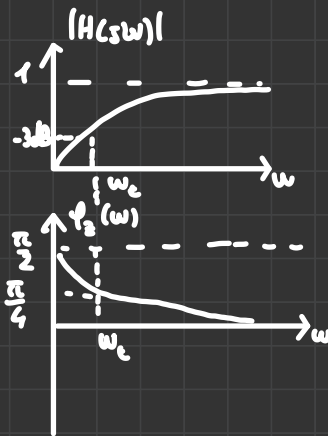
$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 0$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 1$$

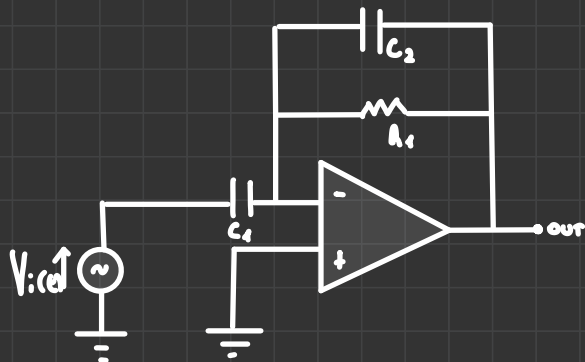
FASE

$$\varphi_2(\omega) = \text{ARCTAN}\left(\frac{\omega}{\omega_c}\right)$$

$$\varphi_2(\omega_c) = \text{ARCTAN}(1) = \frac{\pi}{4}$$



FILTRO PASSA-ALTO ATTIVO DEL 1° ORDINE



$$H(s) = -\frac{C_1}{C_2} \frac{s}{s + \frac{C_1}{C_2}} = \frac{s}{s + \omega_c}$$

$$C_1 = C_2 \quad \omega_c = \frac{C_1}{C_2}$$

FILTRO PASSA-BANDA PASSIVO DEL II ORDINE

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{R}{R + sL + \frac{1}{sC}} = \frac{sCR}{sCR + s^2LC + 1} = \frac{sCR}{LC(s^2 + \frac{1}{LC} + \frac{1}{s})} = \frac{Bs}{(Bs + s^2 + \omega_0^2)}$$

$$s = j\omega$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$H(j\omega) = \frac{Bj\omega}{(Bs^2 - \omega^2 + \omega_0^2)}$$

$$|H(j\omega)| = \frac{\omega B}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\omega B)^2}}$$

$$\varphi(\omega) = \arctan\left(\frac{\omega_0^2 - \omega}{\omega B}\right)$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 0$$

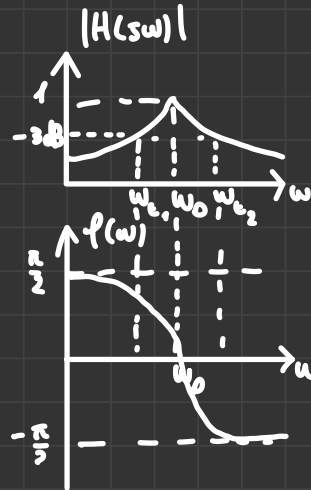
$$\omega = \omega_0 \quad |H(j\omega)|$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 0$$

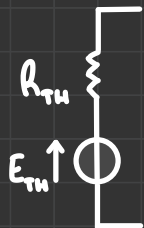
$$\varphi(0) = \frac{\pi}{2}$$

$$\varphi(\omega_0) = 0$$

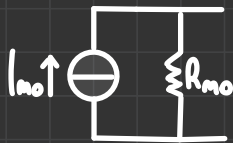
$$\varphi(\infty) = -\frac{\pi}{2}$$



TEOREMA DI THEVENIN E TEOREMA DI NORON



FORMA
THEVENIN



FORMA
NORON

$$E_{TH} = R_{NO} I_{NO}$$

$$I_{NO} = \frac{E_{TH}}{R_{TH}}$$

$$R_{TH} = R_{NO}$$

TEOREMA DEL MASSIMO TRASFERIMENTO DI POTENZA



$$\left\{ \begin{array}{l} P_{\max} = \frac{E_{TH}^2}{4R_{TH}} \\ R_C = R_{eq} = R_{TH} \end{array} \right.$$

$$V = \frac{R_C}{R_C + R_{TH}} \cdot E_{TH}$$

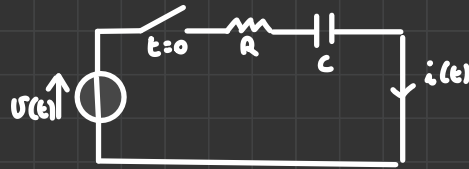
$$i = \frac{V}{R_C} = \frac{E_{TH}}{R_C + R_{TH}}$$

$$P = R_C \cdot i^2 = R_C \cdot \frac{E_{TH}^2}{(R_C + R_{TH})^2} = R_C \cdot \frac{E_{TH}^2}{R_C^2 + R_{TH}^2 + 2R_C R_{TH}} = \frac{E_{TH}^2}{R_C + \frac{R_{TH}^2}{R_C} + 2R_{TH}}$$

$$\frac{d}{dR_C} \left(R_C + \frac{R_{TH}^2}{R_C} + 2R_{TH} \right) = -\frac{R_{TH}^2}{R_C^2} + 1 = 0 \rightarrow R_C = R_{TH}$$

$$P_{\max} = R_{TH} \frac{E_{TH}^2}{(2R_{TH})^2} = \frac{E_{TH}^2}{4R_{TH}}$$

CARICA E SCARICA DEL CONDENSATORE



$$i(t) = \begin{cases} 0 & t \leq 0^- \\ i(t) & t > 0^+ \end{cases}$$

$$e(t) = E_{u_{u_i}}(t)$$

$$e(t) = v_R(t) + v_C(t) \quad t \geq 0^+$$

$$E = R \cdot i + v_C \rightarrow E = R \cdot C \frac{dv_C}{dt} + v_C$$

$$v_C(t) = v_{C_f}(t) + v_{C_p}(t)$$

$$R \cdot C \frac{dv_C(t)}{dt} + v_C(t) = 0$$

$$RC\alpha + 1 = 0 \rightarrow \alpha = -\frac{1}{RC}$$

$$v_{C_f}(t) = k_1 e^{-\frac{1}{RC}t} \quad v_{C_p}(t) = k_2$$

$$v_C(t) = k_1 e^{-\frac{1}{RC}t} + k_2$$

$$\begin{cases} v_C(0) = k_1 + k_2 \\ \lim_{t \rightarrow \infty} v_C(t) = E = k_2 \end{cases}$$

$$\begin{cases} k_1 = v_C(0) - E \\ k_2 = E \end{cases}$$

$$\begin{cases} v_C(t) = (v_C(0) - E) e^{-\frac{1}{RC}t} + E & t \geq 0^+ \\ i(t) = \frac{1}{R} (v_C(0) - E) e^{-\frac{1}{RC}t} & t \geq 0^+ \end{cases}$$

$$|v_C(0)| < |E| \quad \text{CARICA}$$

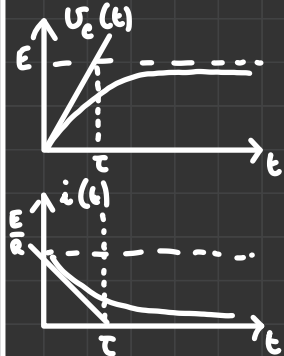
$$|v_C(0)| > |E| \quad \text{SCARICA}$$

$$v_C(0) = 0$$

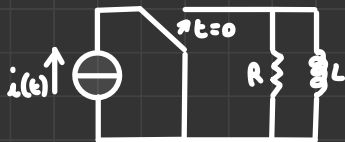
$$v_C(t) = E (1 - e^{-\frac{1}{RC}t})$$

$$i(t) = \frac{E}{R} e^{-\frac{1}{RC}t}$$

$$RC = \tau$$



CARICA E SCARICA DELL'INDUTTORE



$$i(t) = i_{L_g}(t)$$

$$i(t) - Gv(t) - i_L(t) = 0 \quad t \geq 0^+$$

$$I = Gv - i_L \rightarrow I = GL \frac{di_L}{dt} + i_L$$

$$i_L(t) = i_{L_g}(t) + i_{L_p}(t)$$

$$GL \frac{di_L(t)}{dt} + i_L(t) = 0$$

$$GL \alpha + 1 = 0 \rightarrow \alpha = -\frac{1}{GL}$$

$$i_{L_g}(t) = k_1 e^{-\frac{1}{GL}t} \quad i_{L_p}(t) = k_2$$

$$i_L(t) = k_1 e^{-\frac{1}{GL}t} + k_2$$

$$\begin{cases} i_L(0^+) = k_1 + k_2 \\ \lim_{t \rightarrow \infty} i_L(t) = I = k_2 \end{cases}$$

$$\begin{cases} k_1 = i_L(0^+) - I \\ k_2 = I \end{cases}$$

$$\begin{cases} i_L(t) = (i_L(0^+) - I) e^{-\frac{1}{GL}t} + I \\ v(t) = \frac{1}{G} (i_L(0^+) - I) e^{-\frac{1}{GL}t} \end{cases} \quad \begin{matrix} t \geq 0^+ \\ t \geq 0^+ \end{matrix}$$

$$|i_L(0^+)| < I \quad \text{CARICA}$$

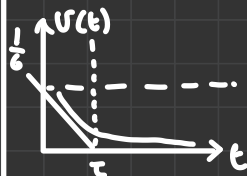
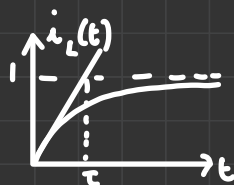
$$|i_L(0^+)| > I \quad \text{SCARICA}$$

$$i_L(0^+) = 0$$

$$i_L(t) = I (1 - e^{-\frac{1}{GL}t})$$

$$v(t) = \frac{1}{G} (e^{-\frac{1}{GL}t})$$

$$GL = \tau$$



CIRCUITI IN REGIME SINUSOIDALE E FASORI

$$\begin{aligned} v(t) &= V_m \sin(\omega t + \gamma) \longrightarrow v(t) = \frac{V_m e^{j(\omega t + \gamma)} - V_m e^{-j(\omega t + \gamma)}}{2j} \\ i(t) &= I_m \sin(\omega t + \beta) \longrightarrow i(t) = \frac{I_m e^{j(\omega t + \beta)} - I_m e^{-j(\omega t + \beta)}}{2j} \end{aligned}$$

FASORI

$$\bar{V} = \frac{V_m e^{j\gamma}}{2j} \quad \bar{I} = \frac{I_m e^{j\beta}}{2j}$$

$$\begin{aligned} v(t) &= \bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t} \\ i(t) &= \bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t} \end{aligned}$$

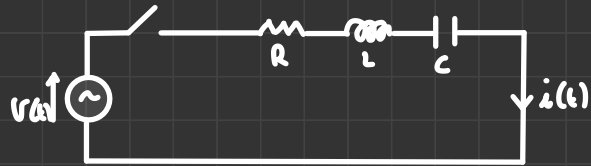
FASORI

$$\bar{V} = V_m e^{j\gamma} \quad \bar{I} = I_m e^{j\beta}$$

$$v(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j}$$

$$i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j}$$

FASORI NEI CIRCUITI RLC CON FORZANTI SINUSOIDALI



$$v(t) = V_m \sin(\omega t + \theta)$$

$$\begin{cases} v_g(t) = R i(t) + L \frac{di(t)}{dt} + v_c(t) \\ i(t) = C \frac{dv_c(t)}{dt} \end{cases}$$

$$v_{gp}(t) = V_m \sin(\omega t + \theta) \quad v_{gp}(0) = V_m \sin(\theta)$$

$$i(t) = I_m \sin(\omega t + \beta)$$

$$v_{gp}(t) = \frac{1}{C} \int_0^t i(t) dx + v_x(0)$$

$$v(t) = \frac{\bar{v} e^{j\omega t} - \bar{v}^* e^{-j\omega t}}{2j}$$

$$i(t) = \frac{\bar{i} e^{j\omega t} - \bar{i}^* e^{-j\omega t}}{2j}$$

$$\bar{V}_g = R \bar{I} + j\omega L \bar{I} + \frac{1}{j\omega C} \bar{I} = (R + j(\omega L - \frac{1}{\omega C})) \bar{I}$$

$$z = R + j(\omega L - \frac{1}{\omega C})$$

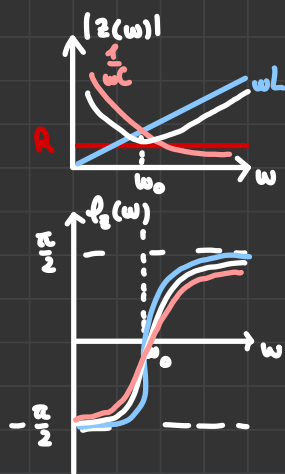
$$\bar{V} = z \bar{I}$$

STUDIO L'IMPIEDENZA AL VARIARE DI ω NEI CIRCUITI RLC SERIE

$$Z(\omega) = R + j(\omega L - \frac{1}{\omega C})$$

$$|Z(\omega)| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

$$\varphi_z(\omega) = \text{ARCTAN}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$



$$\omega < \omega_0$$

OHMICO - CAPACITIVO

$$R=0$$

PUR. CAPACITIVO

$$\omega > \omega_0$$

OHMICO - INDUTTIVO

$$R=0$$

PUR. INDUTTIVO

$$\omega \rightarrow \infty$$

$$\omega L \gg \frac{1}{\omega C}$$

$$\varphi_z(\omega) \rightarrow \frac{\pi}{2}$$

$$\omega = \omega_0$$

$$\varphi_z(\omega) = 0$$

$$\omega \rightarrow 0$$

$$\frac{1}{\omega C} \gg \omega L$$

$$\varphi_z(\omega) \rightarrow -\frac{\pi}{2}$$

POTENZA IN REGIME PERMANENTE SINUSOIDALE

$$v(t) = V_m \sin(\omega t + \alpha) \rightarrow v(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j}$$

$$i(t) = I_m \sin(\omega t + \beta) \rightarrow i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j}$$

$$p(t) = v(t)i(t) = \left(\frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j} \right) \left(\frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j} \right) = -\frac{1}{4} \left(\bar{V} \bar{I} e^{2j\omega t} - \bar{V} \bar{I}^* - \bar{V}^* \bar{I} + (\bar{V} \bar{I})^* e^{-2j\omega t} \right)$$

$$= \underbrace{-\frac{1}{2} \frac{\bar{V} \bar{I} e^{2j\omega t} + \bar{V}^* \bar{I}^* e^{-2j\omega t}}{2}}_{\text{VAR. NEL TEMPO}} + \underbrace{\frac{1}{2} \frac{\bar{V} \bar{I}^* + \bar{V}^* \bar{I}}{2}}_{\text{COSTANTE}}$$

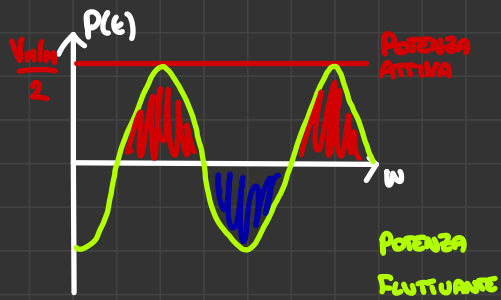
$$\bar{V} \bar{I} = V_m I_m e^{j(\alpha + \beta)} \quad \bar{V} \bar{I}^* = V_m I_m e^{j(\alpha - \beta)}$$

$$(\bar{V} \bar{I})^* = V_m I_m e^{-j(\alpha + \beta)} \quad \bar{V}^* \bar{I} = V_m I_m e^{-j(\alpha - \beta)}$$

$$\alpha - \beta = \varphi \quad \alpha + \beta = 2\alpha - \varphi$$

$$= -\frac{1}{2} V_m I_m \frac{e^{j(2\omega t + 2\alpha - \varphi)}}{2} + e^{-j(2\omega t + 2\alpha - \varphi)} + \frac{1}{2} V_m I_m \frac{e^{j\varphi} + e^{-j\varphi}}{2} = \underbrace{-\frac{1}{2} V_m I_m \cos(2\omega t + 2\alpha - \varphi)} + \underbrace{\frac{1}{2} V_m I_m \cos(\varphi)}$$

$$\left\{ \begin{array}{l} V_{\text{EFF}} = \frac{V_m}{\sqrt{2}} \rightarrow V_m = V_{\text{EFF}} \sqrt{2} \\ I_{\text{EFF}} = \frac{I_m}{\sqrt{2}} \rightarrow I_m = I_{\text{EFF}} \sqrt{2} \end{array} \right. \longrightarrow p(t) = V_{\text{EFF}} I_{\text{EFF}} \cos(\varphi) - V_{\text{EFF}} I_{\text{EFF}} \cos(2\omega t + 2\alpha - \varphi)$$



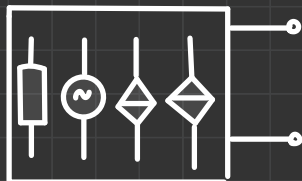
$$\dot{P}_e = \bar{V} \bar{I}^*$$

$$\dot{P}_e = P + jQ$$

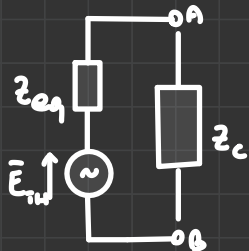
$$P = |\dot{P}_e| \cos(\varphi)$$

$$Q = |\dot{P}_e| \sin(\varphi)$$

TEOREMA DEL MASSIMO TRASFERIMENTO DI POTENZA PER RETI DINAMICHE



$$z_c = z_{eq}^*$$



$$I = \frac{\bar{E}_{tu}}{z_c + z_{eq}} = \frac{\bar{E}_{tu}}{(R_c + R_{eq}) + j(X_c + X_{eq})}$$

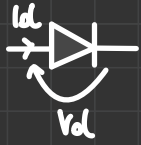
$$P_c = 2 I^2 = R I^2 + j X I^2$$

$$P_c = R I^2 = R \frac{\bar{E}_{tu}^2}{(R_c + R_{eq})^2 + (X_c + X_{eq})^2}$$

$$X_c = -X_{eq} \quad X_{TOT} = 0$$

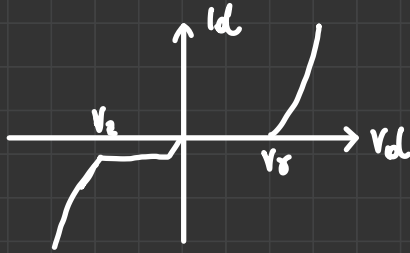
$$z_c = z_{eq}^*$$

DIODI

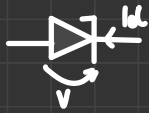


DIRETTA

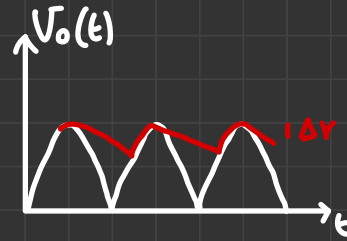
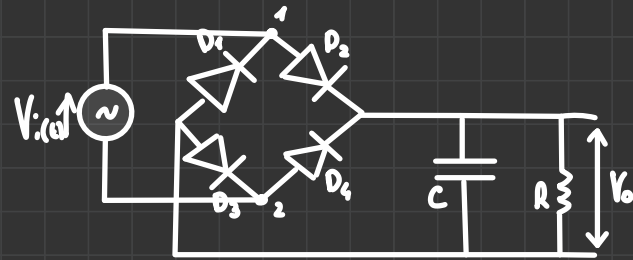
INVERSA



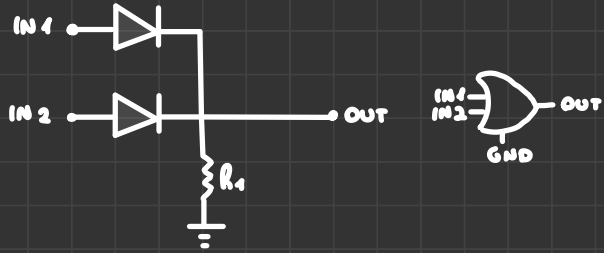
DIODO ZENER



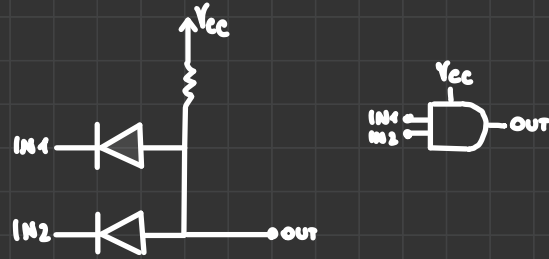
PONTE DI GRATE O RADDOLZITORE A DOPPIA SENONDA



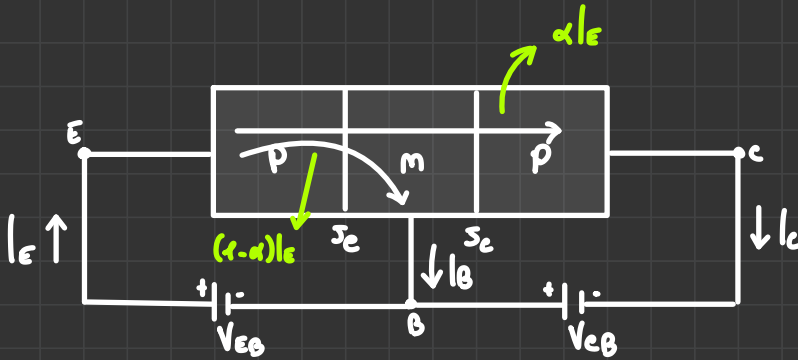
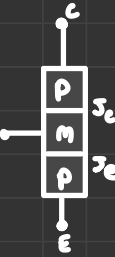
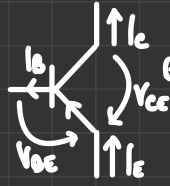
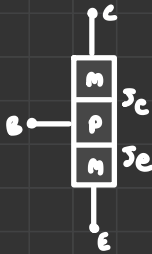
PORTA OR



PORTA AND



TRANSISTOR

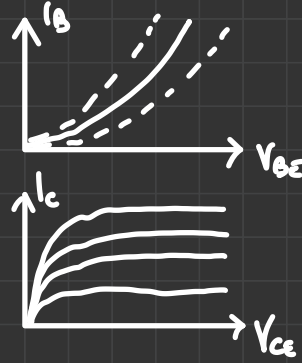
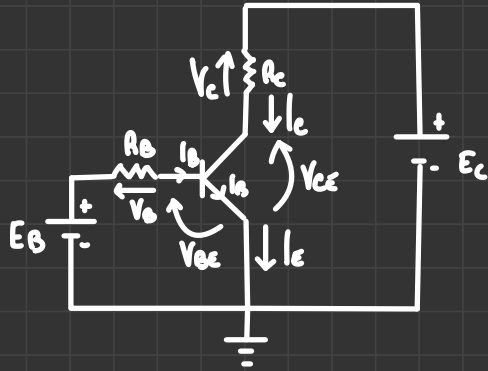


$$I_C = \alpha I_E$$

$$I_B = (1 - \alpha) I_E$$

$$I_E = I_C + I_B$$

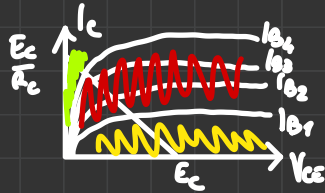
Caratteristiche di Ingresso e di Uscita



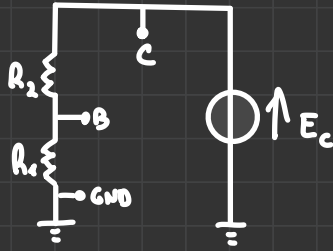
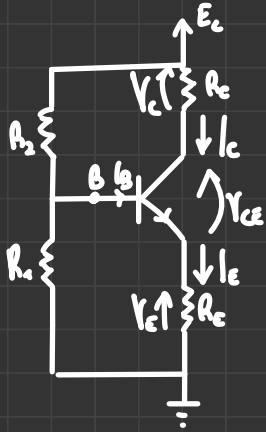
POLARIZZAZIONE

$$E_c = V_c + V_{ce} \rightarrow V_c = E_c - V_{ce} \rightarrow R_c I_c = E_c - V_{ce} \rightarrow I_c = \frac{E_c}{R_c} - \frac{V_{ce}}{R_c}$$

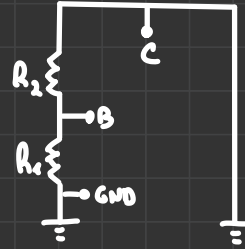
$$\begin{cases} V_{ce} = 0 & I_c = \frac{E_c}{R_c} \\ I_c = 0 & V_{ce} = E_c \end{cases}$$



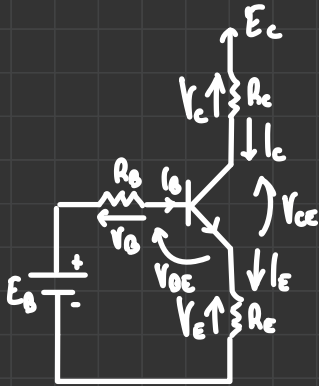
POLARIZZAZIONE CON PARTITORE DI BASE



$$V_{B0} = E_C \frac{R_1}{R_1 + R_2}$$



$$R_{eq} = \frac{R_1 \cdot R_2}{R_2 + R_1}$$



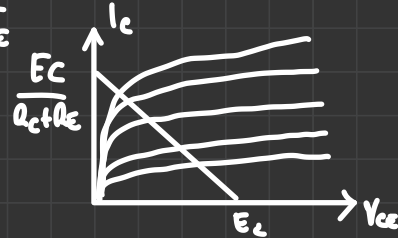
$$E_B = V_B + V_{BE} + V_E$$

$$E_C = V_C + V_{CE} + V_E$$

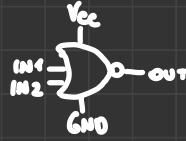
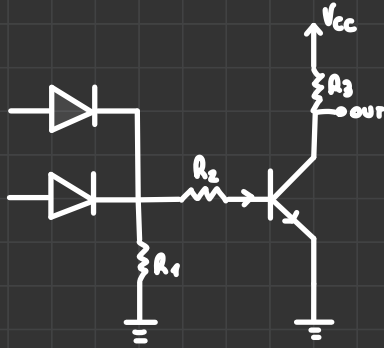
$$I_C = \alpha I_E \approx I_E$$

$$E_C = R_C I_C + V_{CE} + R_E I_E \rightarrow E_C = R_C I_C + V_{CE} + R_E I_C$$

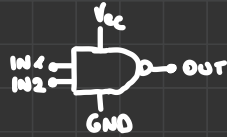
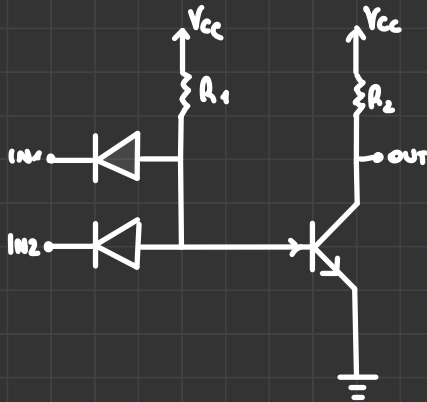
$$I_C = \frac{E_C}{R_C + R_E} - \frac{V_{CE}}{R_C + R_E}$$



PORTA NOR E NAND

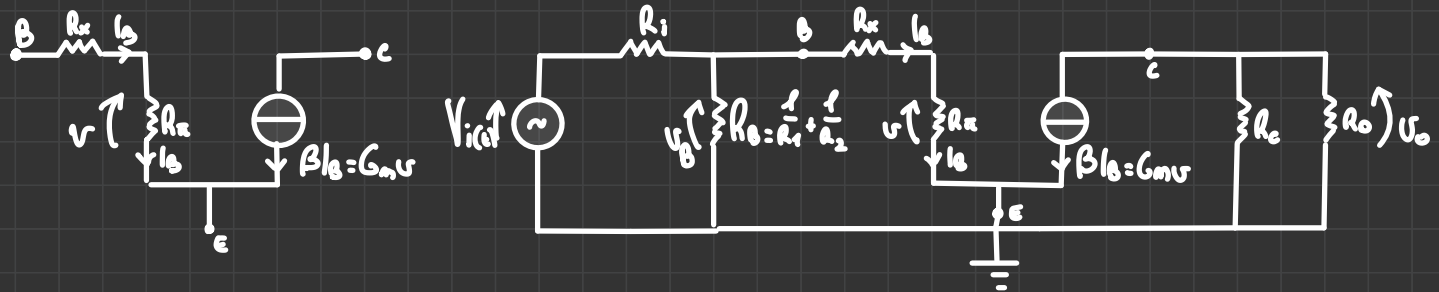
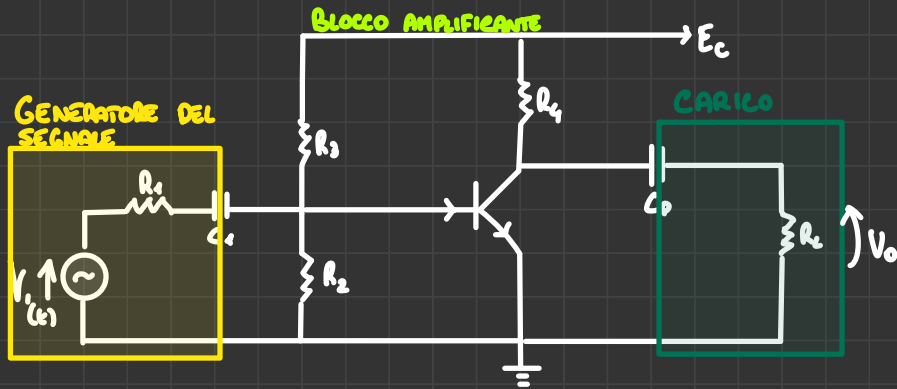


NOR



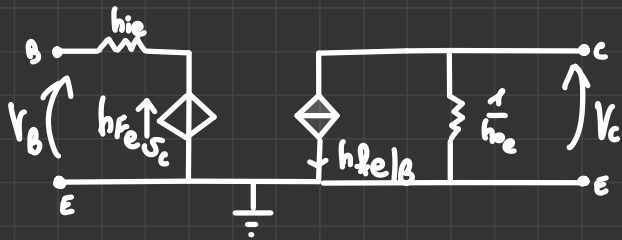
NAND

MODELLI A PARAMETRI |BJT| π |BJT|



$$A = \frac{V_o}{V_i}$$

Modello a Parametri H



$$h_{ie} = R_x + R_{\pi}$$

$$h_{fe} = \beta$$

$$h_{oe} = 0$$

$$h_{re} = 0$$

FILTRO PASSA-BASSO PASSIVO DEL 1° ORDINE

$$H(s) = \frac{1}{R + \frac{1}{sC}} = \frac{1}{sCR + 1} = \frac{1}{RC(s + \frac{1}{RC})}$$

$$s = j\omega$$

$$H(j\omega) = \frac{1}{RC(j\omega + \frac{1}{RC})}$$

$$|H(j\omega)| = \frac{1}{RC \sqrt{\omega^2 + (\frac{1}{RC})^2}}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 1$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 0$$

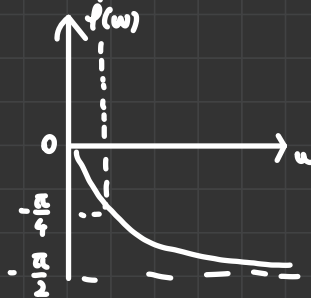
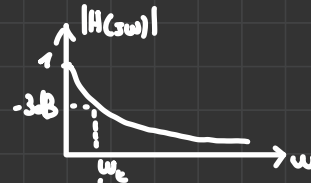
$$\varphi(\omega) = \text{ARCTAN}(-\omega RC)$$

$$|H(j\omega)| = \frac{1}{\sqrt{2}} = -3.01 \text{ dB}$$

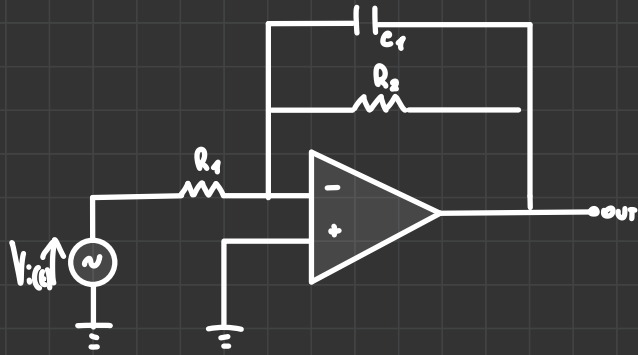
$$\frac{1}{RC \sqrt{\omega_c^2 + (\frac{1}{RC})^2}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c^2 + (\frac{1}{RC})^2 = 2 \left(\frac{1}{RC}\right)^2 \rightarrow \omega_c = \frac{1}{RC}$$

$$H(s) = \frac{\omega_c}{s + \omega_c}$$

$$\varphi(\omega_c) = \text{ARCTAN}(-\omega_c RC) = \text{ARCTAN}(-1) = -\frac{\pi}{4}$$



FILTRO PASSA BASSO ATTIVO DEL 1° ORDINE



$$H(s) = - \frac{G_1}{sC_1 + G_2} = - \frac{\frac{G_1}{C_1}}{s + \frac{G_2}{C_1}} = - \frac{\omega_c}{s + \omega_c}$$
$$\omega_c = \frac{G_1}{C_1} \quad G_1 = G_2$$

FILTRO PASSA-ALTO PASSIVO DEL 1° ORDINE

$$H(s) = 1 - H_{LP}(s) = 1 - \frac{\omega_c}{s + \omega_c} = \frac{s}{s + \omega_c}$$

$$\omega_c = \frac{1}{RC}$$

$$s = j\omega$$

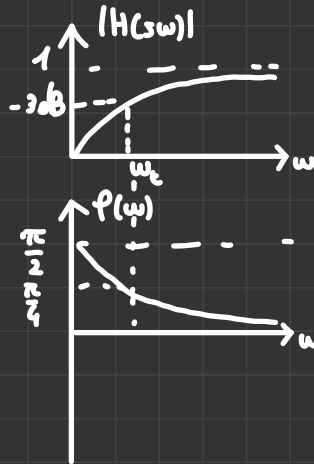
$$H(j\omega) = \frac{j\omega}{j\omega + \omega_c}$$

$$|H(j\omega)| = \frac{\omega^2}{\sqrt{\omega^2 + \omega_c^2}}$$

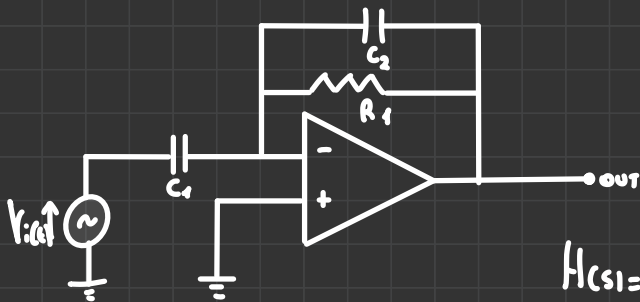
$$\varphi(\omega) = \text{ARCTAN}\left(\frac{\omega_c}{\omega}\right)$$

$$\varphi(\omega_c) = \text{ARCTAN}(1) = \frac{\pi}{4}$$

$$\begin{aligned} \omega \rightarrow 0 & \quad |H(j\omega)| \rightarrow 0 \\ \omega \rightarrow \infty & \quad |H(j\omega)| \rightarrow 1 \end{aligned}$$



FILTRO PASSA ALTO ATTIVO DEL 1° ORDINE



$$H(s) = -\frac{C_1}{C_2} \frac{s}{s + \frac{1}{R_1 C_2}} = -\frac{s}{s + \omega_c}$$

$$\omega_c = \frac{1}{R_1 C_2} \quad C_1 = C_2$$

FILTRO PASSA-ALTO PASSIVO DEL II ORDINE

$$H(s) = \frac{sL}{R + sL + \frac{1}{sC}} = \frac{s^2 LC}{s^2 LC + s + \frac{1}{LC}} = \frac{s^2 LC}{LC(s^2 + s\frac{R}{L} + \frac{1}{LC})} = \frac{s^2}{\frac{R}{L}s + s^2 + \frac{1}{LC}} = \frac{s^2}{Bs + s^2 + \omega_0^2}$$

$$s = j\omega$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$H(j\omega) = \frac{-\omega^2}{Bs + \omega_0^2 - \omega^2}$$

$$|H(j\omega)| = \frac{\omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (B\omega)^2}}$$

$$\varphi(\omega) = \begin{cases} \arctan\left(-\frac{\omega B}{\omega_0^2 - \omega^2}\right) + \pi & \omega \leq \omega_0 \\ \arctan\left(-\frac{\omega B}{\omega_0^2 - \omega^2}\right) & \omega > \omega_0 \end{cases}$$

$$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 0$$

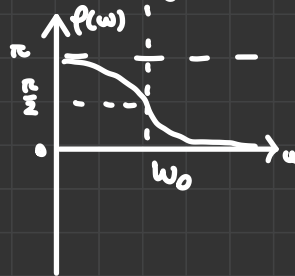
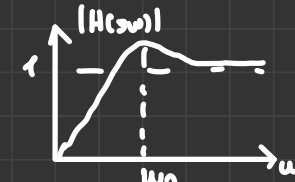
$$\omega = \omega_0 \quad H(j\omega)$$

$$\omega \rightarrow \infty \quad H(j\omega) \rightarrow 1$$

$$\varphi(0) = \pi$$

$$\varphi(\omega_0) = -\frac{\pi}{2} + \pi = \frac{\pi}{2}$$

$$\varphi(\infty) = 0$$



FILTRO PASSO-BANDA PASSIVO DEL II ORDINE

$$H(s) = \frac{R}{R+sL+\frac{1}{sC}} = \frac{sCR}{sCR+s^2LC+1} = \frac{sCR}{LC(s^2+s^2+\frac{1}{LC})} = \frac{b_s}{b_s+s^2+\omega_0^2}$$

$$\omega_0 = \sqrt{\frac{1}{LC}} \quad b = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{b j\omega}{b j\omega + \omega_0^2 - \omega^2}$$

$$|H(j\omega)| = \frac{\omega b}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\omega b)^2}}$$

$$\varphi(\omega) = \arctan\left(\frac{\omega_0^2 - \omega^2}{\omega b}\right)$$

$$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 0$$

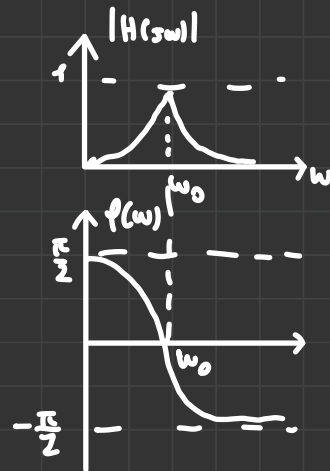
$$\omega = \omega_0 \quad H(j\omega) \rightarrow 1$$

$$\omega \rightarrow \infty \quad H(j\omega) \rightarrow 0$$

$$\varphi(0) = \frac{\pi}{2}$$

$$\varphi(\omega_0) = 0$$

$$\varphi(\infty) = -\frac{\pi}{2}$$



FILTRO ELIMINA BANDA PASIVO DEL II ORDINE

$$H(s) = \frac{sL + \frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{s^2LC + 1}{sCR + s^2LC + 1} = \frac{LC(s^2 + \frac{1}{LC})}{LC(\frac{R}{L}s + s^2 + \frac{1}{LC})} = \frac{s^2 + \omega_0^2}{Bs + s^2 + \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \theta = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{\omega_0^2 - \omega^2}{Bs + \omega_0^2 - \omega^2}$$

$$|H(j\omega)| = \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\omega\theta)^2}}$$

$$\phi(\omega) = \arctan\left(-\frac{\omega\theta}{\omega_0^2 - \omega^2}\right)$$

$$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 1$$

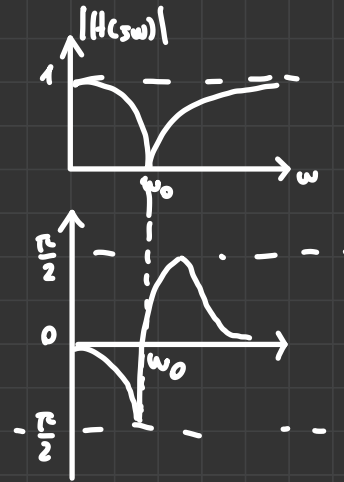
$$\omega = \omega_0 \quad H(j\omega) \rightarrow 0$$

$$\omega \rightarrow \infty \quad H(j\omega) \rightarrow 1$$

$$\phi(0) = 0$$

$$\phi(\omega_0) = -\frac{\pi}{2}$$

$$\phi(\infty) = 0$$



FILTRO PASSA-BASSO PASSIVO DEL II ORDINE

$$H(s) = \frac{1}{s^2 LC + \frac{L}{s}} = \frac{1}{s^2 LC + \frac{1}{s}} = \frac{1}{LC(s^2 + \frac{1}{LC})} = \frac{\omega_0^2}{s^2 + \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \beta = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{\omega_0^2}{\beta j\omega + \omega_0^2 - \omega^2}$$

$$|H(j\omega)| = \frac{\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\omega\beta)^2}}$$

$$\phi(\omega) = \begin{cases} \text{ARCTAN}\left(-\frac{\omega\beta}{\omega_0^2 - \omega^2}\right) & \omega \leq \omega_0 \\ \text{ARCTAN}\left(-\frac{\omega\beta}{\omega_0^2 - \omega^2}\right) - \pi & \omega > \omega_0 \end{cases}$$

$$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 1$$

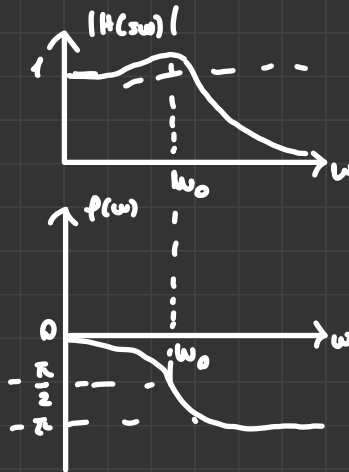
$$\omega = \omega_0$$

$$\omega \rightarrow \infty \quad H(j\omega) \rightarrow 0$$

$$\phi(0) = 0$$

$$\phi(\omega_0) = -\frac{\pi}{2}$$

$$\phi(\infty) = -\pi$$



FILTRO PASSA BANDA ATIVO DEL 11 ORDINE

